
UNIVERSITY OF KALYANI



Statistical Analysis of Option Pricing

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Declearation

Myself, **Ritabrata Sadhukhan** and **Rupak Chanda**, student of B.Sc. in Statistics, hereby declare that this dissertation titled

”Statistical Analysis of Option Pricing”

submitted by us to the Department of Statistics, Kalyani Mahavidyalaya, is a genuine work carried out by me under the guidance and supervision of Prof. Kakali Dutta and Prof. Kamal Krishna Khanda at the Department of Statistics, Kalyani Mahavidyalaya, India submitted in partial fulfillment of the requirements for the degree of Bachelor of Science (B.Sc.) in Statistics. We also affirm that no part of the project uses unacknowledged materials and resources.

We further declare that this work has not been submitted anywhere else for the award of any other degree or diploma.

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Contents

1	Abstract	9
2	Introduction to Option Pricing	10
2.1	Overview of Derivatives and option	10
2.1.1	What are Derivatives?	10
2.1.2	Understanding Options	10
2.1.3	Option Pricing Models	10
2.1.4	Price Influencing Factors	11
2.1.5	Volatility Patterns	12
2.1.6	Risk Measures	12
2.1.7	Conclusion	12
2.2	Historical Development: From Black-Scholes to Advanced Models	12
2.3	Importance of Non-Standard Assumptions in Modern Markets	14
2.3.1	Introduction: Beyond Idealized Frameworks	14
2.3.2	Stochastic Volatility Modeling	15
2.3.3	Discontinuous Asset Price Movements	15
2.3.4	Stochastic Interest Rate Dynamics	15
2.3.5	Volatility Surface Modeling	15
2.3.6	Market Microstructure Considerations	16
2.3.7	Asymmetric Volatility Response	16
2.3.8	Heavy-Tailed Return Distributions	16
2.3.9	Conclusion: Theoretical Rigor and Empirical Validation	16
2.4	Objectives of the Module	17
3	The Black-Scholes Model	18
3.1	Assumptions Underlying the Black-Scholes Model	18
3.1.1	Constant Volatility Assumption	18
3.1.2	Static Interest Rate Framework	18
3.1.3	Dividend Exclusion	18
3.1.4	Frictionless Market Premise	18
3.1.5	Perfect Liquidity Condition	18
3.1.6	Efficient Market Hypothesis	18
3.1.7	Geometric Brownian Motion	18
3.1.8	European Exercise Constraint	19
3.1.9	Log-Normal Return Distribution	19
3.1.10	Theoretical Value and Practical Limitations	19
3.1.11	Rationale for the Model Assumptions	19
3.1.12	Conclusion	19
3.2	Derivation Using Stochastic Calculus	19
3.2.1	Step 1: Modeling Asset Price Dynamics	20
3.2.2	Step 2: Itô's Lemma Application	20
3.2.3	Step 3: Constructing the Hedged Portfolio	20
3.2.4	Step 4: Portfolio Dynamics Analysis	20
3.2.5	Step 5: No-Arbitrage Principle	20
3.2.6	Step 6: Black-Scholes PDE	20
3.2.7	Boundary Conditions	21
3.2.8	Interpretation and Significance	21
3.2.9	Conclusion	21
3.3	Analytical Solution for European Call and Put Options	21
3.3.1	Key Notation	21
3.3.2	European Call Option Formula	21
3.3.3	European Put Option Formula	21
3.3.4	Derivation of d_1 and d_2 Parameters	22
3.3.5	Interpretation of Components	23
3.3.6	Put-Call Parity Relationship	23

3.3.7	Practical Applications	23
3.3.8	Model Limitations	23
3.3.9	Summary of Formulas	23
3.3.10	Conclusion	23
3.4	Determination of parameters with Crank Nicolson Method	24
3.4.1	Choose an Objective Function	24
3.4.2	Embedded Crank–Nicolson Solver	24
3.4.3	Optimization Algorithm	24
3.4.4	Regularization & Parameter Bounds	24
3.4.5	Diagnostic & Convergence Checks	24
3.4.6	Implied Volatility Extraction	24
3.4.7	Python Code for Estimate Parameters	25
3.4.8	Implementation	25
3.5	Limitations: Constant Volatility and Lognormal Assumption	26
3.5.1	The Constant Volatility Assumption	26
3.5.2	The Lognormal Price Distribution Assumption	27
3.5.3	Practical Implications for Risk Management	27
3.5.4	Advanced Modeling Approaches	27
3.5.5	Implementation Considerations	27
3.5.6	Conclusion	27
4	Monte Carlo Simulation under Geometric Brownian Motion	28
4.1	Euler–Maruyama Discretization Scheme	28
4.1.1	Stochastic Differential Equation (SDE)	28
4.1.2	Euler–Maruyama Approximation	28
4.1.3	Euler–Maruyama Applied to GBM (Geometric Brownian Motion)	28
4.1.4	Advantages and Limitations	29
4.1.5	Higher-Order Alternatives	29
4.1.6	Conclusion	29
4.2	Simulating Multiple Price Paths	29
4.2.1	Simulation of GBM with the help of Euler-Maruyama	29
4.2.2	Simulation of Asset Price Using Euler–Maruyama Method	30
4.3	Option Pricing via Monte Carlo Integration	31
4.3.1	Using Euler-Maruyama	31
4.3.2	Risk-Neutral Valuation as an Expectation	31
4.3.3	Simulation of Underlying Price Paths	32
4.3.4	Payoff Estimation and Discounting	32
4.3.5	Convergence and Error	32
4.3.6	Variance Reduction Techniques	32
4.3.7	Greeks via Monte Carlo	32
4.3.8	Practical Considerations	32
5	Stochastic Volatility Models – The Heston Framework	33
5.1	Stochastic Volatility Models – The Heston Framework	33
5.1.1	Introduction to Stochastic Volatility	33
5.1.2	The Heston Model	33
5.1.3	Why the Heston Model?	33
5.1.4	Parameter Calibration	33
5.1.5	Closed-Form Solution for European Options	33
5.1.6	Advantages	34
5.1.7	Limitations	34
5.1.8	Extensions and Applications	34
5.1.9	Conclusion	34
5.2	The Heston Model: Formulation and Assumptions	35
5.3	Simulation Techniques (Euler/Milstein for Correlated SDEs)	36
5.3.1	Correlated Increment Generation	37
5.3.2	Euler–Maruyama Method	37
5.3.3	Milstein Method	38

5.3.4	Higher-Order and Alternative Schemes	38
5.3.5	Practical Implementation Considerations	38
5.3.6	Comparison of Euler–Maruyama and Milstein	38
5.3.7	Conclusion	38
5.4	Pricing via Monte Carlo under Stochastic Volatility	39
5.4.1	Model Specification	39
5.4.2	Time Discretization	39
5.4.3	Path Generation	39
5.4.4	Payoff and Estimation	39
5.4.5	Variance Reduction	39
5.4.6	Estimation	39
5.4.7	Convergence Monitoring	39
5.4.8	Practical Considerations	39
5.4.9	Summary	39
5.5	Behavior of Prices under Different Volatility Regimes	40
5.5.1	Regime Detection Mechanisms	40
5.5.2	Impact on Option Pricing	40
5.5.3	Model Calibration and Parameter Estimation	40
5.5.4	Risk Management Implications	40
5.5.5	Portfolio Allocation and Strategic Decisions	40
5.5.6	Empirical Validation and Backtesting	40
6	Calibration and Statistical Inference	41
6.1	Overview of Parameter Estimation in Financial Models	41
6.1.1	Data and Model	41
6.1.2	Log-Likelihood $\rightarrow$ Objective	41
6.1.3	Least-Squares Formulation	41
6.1.4	Optimization	41
6.1.5	Summary	41
6.1.6	Error Estimation of Model	42
6.2	Objective Function: Error Minimization between Market and Model	42
6.2.1	Pricing Error	42
6.2.2	Sum of Squared Errors (SSE)	43
6.2.3	Weighted SSE (Relative Fit)	43
6.2.4	Mean Absolute Error (MAE)	43
6.2.5	Huber Loss (Hybrid Robustness)	43
6.2.6	Regularization (Stability Control)	43
6.2.7	Out-of-Sample Validation	43
6.2.8	Optimization Algorithms	43
6.3	Optimization Methods (Nelder-Mead / L-BFGS-B / MLE)	44
6.3.1	Nelder–Mead Simplex Method	44
6.3.2	L-BFGS-B (Limited-memory BFGS with Bounds)	44
6.3.3	Maximum Likelihood Estimation (MLE)	44
6.3.4	Comparison of Methods	45
7	Comparative Analysis	46
7.1	Pricing Accuracy: Black-Scholes vs. Heston	46
7.1.1	Price Accuracy Comparison	47
7.2	Volatility Surface Fit and the Smile Effect	47
7.2.1	Volatility Surface Fit	47
7.2.2	Smile Effect	48
7.2.3	Conclusion	48
7.3	RMSE, MAPE and Other Statistical Measures of Fit	48
7.3.1	Root Mean Squared Error (RMSE)	48
7.3.2	Mean Absolute Percentage Error (MAPE)	49
7.3.3	Other Common Measures of Fit	49
7.4	Graphical Representations: Error Bands and Surface Plots	49
7.4.1	Error Bands	49

7.4.2	Surface Plot	50
7.5	Interpretation and Inference	50
7.5.1	GBM Error Bands	50
7.5.2	Synthetic Implied-Volatility Surface	51
7.5.3	Bringing both plots together	51
8	Research Contributions and Literature Review	52
8.1	Key Papers on Option Pricing Beyond Black-Scholes	52
8.1.1	Hull and White (1987): Pricing Options on Assets with Stochastic Volatility . . .	52
8.1.2	Heston (1993): A Closed-Form Solution for Options with Stochastic Volatility . . .	53
8.1.3	Bakshi, Cao, Chen (1997): Empirical Performance of Alternative Option Pricing Models	55
8.1.4	Gatheral (2006): The Volatility Surface	57
9	Conclusion and Insights	60
9.1	Summary of Theoretical and Empirical Advancements	60
9.2	Projection of Future	61
10	Appendix	63
10.1	Python Codes	63
11	Bibliography	64
11.1	Academic Articles	64
11.2	Online Resources	64

1 Abstract

- **The Symbiotic Bond: Statistics as the Foundation of Quantitative Finance**

Quantitative finance, the discipline that applies mathematical models and computational techniques to financial markets, often appears as a complex realm of intricate formulas, high-frequency algorithms, and sophisticated derivatives pricing. Yet, beneath this technological veneer lies a fundamental and indispensable bedrock: *statistics*. The relationship between quantitative finance and statistics is not merely one of occasional application; it is a profound symbiosis where statistics provides the essential language, tools, and philosophical framework for understanding, modeling, and managing financial phenomena.

- **Modeling Asset Returns: The Stochastic Heartbeat of Markets**

Financial markets are inherently uncertain and dynamic. Statistics provides the framework for characterizing this uncertainty through probability distributions. Measures such as *expected return* (mean) and *risk* (variance or standard deviation) form the cornerstone of Modern Portfolio Theory (MPT). Time series analysis models phenomena like volatility clustering using ARCH/GARCH frameworks.

- **Derivatives Pricing: Probabilistic Valuation**

The Black–Scholes–Merton model and its successors treat an asset’s price as a stochastic process (e.g., Geometric Brownian Motion). Derivative prices become discounted expectations of pay-offs under a risk-neutral measure. Monte Carlo simulation—a statistical technique for estimating complex integrals via random sampling—is essential when closed-form solutions are unavailable.

- **Risk Management: Quantifying the Unquantifiable**

Value at Risk (VaR) and Expected Shortfall (ES/CVaR) are statistical risk measures that estimate potential losses over a horizon at a confidence level. Computing these metrics involves historical simulation, parametric methods, or Monte Carlo—all rooted in statistical inference. Credit risk models, predicting default likelihoods, use logistic regression and other classification techniques.

- **Algorithmic and High-Frequency Trading: Statistical Arbitrage and Signal Generation**

Statistical arbitrage exploits statistically significant deviations from theoretical price relationships (e.g., pairs trading). Identifying opportunities relies on cointegration and mean-reversion tests. Signal generation uses regression models and pattern-recognition algorithms to separate genuine signals from noise, and backtesting employs statistical performance metrics (Sharpe ratio, Sortino ratio) to mitigate overfitting.

- **Model Calibration and Validation: Ensuring Relevance**

Calibration estimates model parameters from market data via maximum likelihood or method of moments. Validation tests predictive power on out-of-sample data. Hypothesis tests check assumptions (normality of residuals, independence), and goodness-of-fit tests (Kolmogorov–Smirnov, Chi-square) assess distributional choices.

Limitations and the Role of Judgment Statistics, while indispensable, is not a crystal ball. Markets are complex adaptive systems influenced by human behavior, unforeseen “black swan” events, and structural shifts. Statistical measures like correlation can be unstable, and models built on historical data may fail when assumptions break down. Hence, quantitative finance must blend rigorous statistical modeling with economic reasoning, market intuition, and an acute awareness of model limitations.

Finally, Quantitative finance is, in essence, applied statistics operating within the domain of financial markets. From characterizing asset returns to pricing derivatives, from managing multifaceted risks to driving automated trading strategies, and from calibrating to validating models, statistics transforms market uncertainty into quantifiable probabilities and risks. While not infallible, the deep integration of statistical principles endows quantitative finance with rigor, objectivity, and analytical power—qualities essential for navigating an inherently uncertain world.

2 Introduction to Option Pricing

2.1 Overview of Derivatives and option

2.1.1 What are Derivatives?

- Financial tools that derive value from underlying assets (stocks, bonds, commodities)
- Used for:
 - Protecting investments (**hedging**)
 - Making speculative plays
 - Exploiting price differences (**arbitrage**)
- **Common Types:**
 - **Forwards:** Custom OTC contracts for future transactions
 - **Futures:** Standardized exchange-traded contracts
 - **Options:** Right (not obligation) to buy/sell assets
 - **Swaps:** Cash flow exchanges (interest rates/currencies)

2.1.2 Understanding Options

- Contracts providing flexibility through premium payment
- **Two Main Types:**
 - **Calls:** Right to *buy* at strike price
 - **Puts:** Right to *sell* at strike price
- **Key Terminology:**
 - **Strike Price:** Agreed transaction price
 - **Premium:** Option's upfront cost
 - **Expiration:** Contract deadline
 - **In-the-Money (ITM):** Currently profitable position
 - **Out-of-the-Money (OTM):** Not currently profitable
 - **At-the-Money (ATM):** Strike = current price

2.1.3 Option Pricing Models

- **Black-Scholes Model:**
 - The Black-Scholes model originated from the need to fairly and consistently price European-style options in financial markets. Here's a discussion on how the idea to apply the Black-Scholes framework emerged:
 - Historical Motivation Behind Black-Scholes
 1. Need for a Rational Pricing Formula: Prior to Black-Scholes (1973), option pricing lacked a rigorous, universally accepted model. Prices were based on intuition, rules of thumb, or empirical relationships, leading to inconsistencies.
 2. Financial Innovation in the 1960s–70s: The rapid growth of options markets (notably with the creation of the Chicago Board Options Exchange in 1973) demanded a scientific method for pricing and hedging derivative products.
 3. Link to Physics and Brownian Motion: Louis Bachelier (1900) initially proposed modeling stock prices using Brownian motion. Later, economists like Paul Samuelson adopted ****geometric Brownian motion**** to model stock dynamics, leading toward a stochastic process framework.

-
- The Insight: Replication + No Arbitrage

The core insight behind Black-Scholes is:

If you can continuously hedge an option using the underlying asset and a risk-free bond, you can replicate its payoff, hence the price of the option must be equal to the cost of replication.

This leads to:

- * Delta-hedging: Constructing a portfolio that eliminates risk by adjusting the position in the underlying.
 - * Risk-neutral valuation: Pricing based on expected payoff under a measure where all assets grow at the risk-free rate.
- Mathematical Assumptions Used as highlighted in your image:
 - * Constant volatility: σ does not vary with time or asset price.
 - * No dividends: The asset does not pay intermediate cash flows.
 - * Frictionless markets: No transaction costs, continuous trading is possible.
 - * Lognormal distribution: Asset prices follow a geometric Brownian motion:

$$dS = \mu S dt + \sigma S dW_t$$

These allowed the derivation of a partial differential equation (PDE) whose solution gives the price of an option.

- Impact and Why It's Used
 - * It offered the first closed-form solution for option prices.
 - * Provided a benchmark against which all other pricing models are compared.
 - * Enabled the growth of dynamic hedging strategies and structured products.
 - * Easily implemented, hence adopted widely in trading desks, finance textbooks, and research.
- Standard for European options
- Call option formula:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

- Key assumptions:
 - * Constant volatility
 - * No dividends
 - * Frictionless markets

- **Heston Model:**

- Accounts for changing volatility
- Explains volatility smile/skew patterns

2.1.4 Price Influencing Factors

- **Underlying Price:**

- Calls $\uparrow$ when asset $\uparrow$
- Puts $\downarrow$ when asset $\uparrow$

- **Strike Price:** Proximity to current price

- **Time Decay:** Premiums erode as expiration approaches

-
- **Volatility:** Higher uncertainty = higher premiums
 - **Interest Rates:**
 - Higher rates help calls
 - Higher rates hurt puts
 - **Dividends:**
 - Reduce call values
 - Increase put values

2.1.5 Volatility Patterns

- **Volatility Smile:**
 - Higher implied volatility for ITM/OTM options
 - U-shaped curve across strikes
- **Volatility Skew:**
 - Asymmetric volatility patterns
 - OTM puts > OTM calls in volatility

2.1.6 Risk Measures

- **Delta (Δ):** Price sensitivity to asset moves
- **Gamma (Γ):** Delta's sensitivity to price changes
- **Theta (Θ):** Daily time decay (negative value)
- **Vega (ν):** Sensitivity to volatility changes
- **Rho (ρ):** Interest rate sensitivity

2.1.7 Conclusion

- Options derive value from multiple complex factors
- Black-Scholes provides foundation, Heston adds realism
- Greeks enable precise risk management
- Volatility patterns reveal market expectations

2.2 Historical Development: From Black-Scholes to Advanced Models

How mathematicians transformed market chaos into quantifiable risk

1. Pre-1973: The Pioneering Era (1900s-1960s)

- **Louis Bachelier (1900):**
 - Created first option model in thesis *Théorie de la spéculation*
 - Modeled stocks with arithmetic Brownian motion
 - Revolutionary insight: Applied physics to finance
- **Early No-Arbitrage Concepts:**
 - Established "no free lunch" price boundaries
 - Laid groundwork for risk-neutral valuation

2. 1973: The Black-Scholes-Merton Revolution

- **Black & Scholes (1973):**

- First practical European option formula
- Assumptions: Frictionless markets, constant volatility
- Famous equation:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

- **Merton's Contributions:**

- Added dividend yield handling
- Formalized risk-neutral math (martingales)
- Introduced "implied volatility" concept

- **Global Impact:**

- Proved risk elimination via dynamic hedging
- Transformed derivatives into legitimate instruments

3. 1970s-1980s: Adding Realism

- **Merton's Jump-Diffusion (1976):**

- Added market crash/spike modeling
- Captured "black swan" events

- **Garman-Kohlhagen (1983):**

- Currency option adaptation
- Key innovation: Dual interest rates

- **CEV(Constant Elasticity of Variance model) Model:**

- Volatility responds to price levels
- Explained crash vs. rally behavior

4. 1990s: Volatility Breakthroughs

- **Local Volatility (1994):**

- Market-driven approach (Dupire/Derman-Kani)
- Extract volatility surface from option prices

- **Heston Model (1993):**

- Stochastic volatility with mean reversion
- Solved volatility smile problem
- Enabled fast Fourier pricing

5. Late 1990s-2000s: Jump Models

- **Variance Gamma & CGMY:**

- Pure jump models (Lévy processes)
- Captured fat tails and distribution skew

- **SVJ Models:**

- Combined jumps with stochastic volatility
- Crisis-period realism

6. 2000s: Market Standardization

- **SABR Model (2002):**
 - Industry standard for rate options
 - Simple 4-parameter volatility approximation

7. 2010s-Present: Rough Volatility

- **Empirical Discovery:**
 - Volatility has fractal structure ($H < 0.5$)
- **Rough Models:**
 - Fractional Brownian motion integration
 - Perfect volatility term structure fit

8. Machine Learning Frontier

- **Neural Networks:**
 - Learn pricing directly from market data
 - Math-free calibration
- **Gaussian Processes:**
 - Non-parametric volatility surfaces
 - Built-in uncertainty measures

Conclusion

- **Evolution:** Physics $\rightarrow$ Math $\rightarrow$ AI
- **Core Tension:** Elegance vs. realism
- **Next Frontier:** Quantum finance

2.3 Importance of Non-Standard Assumptions in Modern Markets

This analysis examines the necessity of non-standard assumptions in financial modeling, contrasting classical frameworks with contemporary approaches. Where traditional models presuppose market idealizations, modern methodologies incorporate empirical realities such as stochastic volatility, discontinuous price movements, and transaction costs. These advancements significantly enhance pricing accuracy, risk assessment, and strategic decision-making in complex financial environments.

2.3.1 Introduction: Beyond Idealized Frameworks

The Black-Scholes model and its derivatives established foundational principles for option pricing under constrained assumptions. Contemporary financial markets, characterized by heightened volatility and structural complexity, necessitate models incorporating non-standard assumptions that better reflect empirical market behaviors. This project systematically examines seven critical enhancements to classical frameworks and their practical implications.

2.3.2 Stochastic Volatility Modeling

- **Limitation of classical approach:** Presumes constant volatility
- **Empirical reality:** Volatility exhibits time-varying characteristics influenced by market sentiment and macroeconomic factors
- **Modeling advancement:** Heston framework and extensions incorporating mean-reverting variance processes
- **Practical significance:**
 - Accurately replicates volatility smile/skew phenomena
 - Enables dynamic hedging during volatility regime shifts
 - *Market evidence:* VIX term structure dynamics during 2020 market dislocation

2.3.3 Discontinuous Asset Price Movements

- **Limitation of classical approach:** Assumes continuous price paths
- **Empirical reality:** Asset prices experience discontinuous jumps following significant information events
- **Modeling advancement:** Merton jump-diffusion and exponential Lévy processes
- **Practical significance:**
 - Captures leptokurtic return distributions and tail risk
 - Improves pricing of deep out-of-the-money options
 - *Market evidence:* Single-day S&P 500 declines exceeding 5% during crisis periods

2.3.4 Stochastic Interest Rate Dynamics

- **Limitation of classical approach:** Static interest rate assumption
- **Empirical reality:** Rates exhibit path-dependent stochastic behavior
- **Modeling advancement:** Cox-Ingersoll-Ross and Hull-White interest rate processes
- **Practical significance:**
 - Essential for interest rate derivative valuation
 - Accurate modeling of forward rate curves
 - *Market evidence:* Yield curve inversions preceding economic recessions

2.3.5 Volatility Surface Modeling

- **Limitation of classical approach:** Flat volatility assumption
- **Empirical reality:** Implied volatility varies by strike and maturity
- **Modeling advancement:** Dupire local volatility and stochastic volatility surfaces
- **Practical significance:**
 - Consistent pricing across option moneyness dimensions
 - Reveals market-implied probability distributions
 - *Market evidence:* Persistent volatility skew in equity index options

2.3.6 Market Microstructure Considerations

- **Limitation of classical approach:** Frictionless market assumption
- **Empirical reality:** Transaction costs and liquidity constraints impact execution
- **Modeling advancement:** Transaction cost models incorporating bid-ask spreads and market impact
- **Practical significance:**
 - Realistic assessment of arbitrage boundaries
 - Performance attribution for trading strategies
 - *Market evidence:* Liquidity premiums in crisis periods

2.3.7 Asymmetric Volatility Response

- **Limitation of classical approach:** Symmetric return distribution
- **Empirical reality:** Volatility increases more during market declines (leverage effect)
- **Modeling advancement:** GARCH-type processes with asymmetric volatility clustering
- **Practical significance:**
 - Accurate risk modeling during market stress
 - Improved estimation of value-at-risk (VaR)
 - *Market evidence:* VIX-Equity correlation dynamics

2.3.8 Heavy-Tailed Return Distributions

- **Limitation of classical approach:** Normality assumption
- **Empirical reality:** Excess kurtosis in asset returns
- **Modeling advancement:** Variance Gamma and tempered stable processes
- **Practical significance:**
 - Proper pricing of catastrophe options risk hedging strategies
 - *Market evidence:* Frequency of 4+ standard deviation market moves

2.3.9 Conclusion: Theoretical Rigor and Empirical Validation

- Classical models provide conceptual frameworks but insufficiently capture market complexities
- Non-standard assumptions bridge theoretical finance with empirical market structure
- Contemporary models demonstrate superior performance during stress periods
- Future research should integrate behavioral finance factors and machine learning techniques

2.4 Objectives of the Module

The primary objective of this project is to evaluate and compare the performance of two widely used option pricing models—the classical Black-Scholes model and the Heston stochastic volatility model—by estimating their parameters and assessing their pricing accuracy on real market data. The study involves:

Estimating model parameters using numerical and analytical methods, including the Crank–Nicolson finite difference scheme for the Black-Scholes PDE, and optimization-based calibration for the Heston model.

Implementing Monte Carlo simulation techniques to generate option prices under both models.

Quantitatively evaluating model performance by calculating and minimizing pricing errors using **Mean Absolute Percentage Error (MAPE)** and **Root Mean Squared Error (RMSE)**.

Visualizing volatility smiles and error surfaces to better understand model behavior under different market conditions.

This comparative analysis aims to identify the practical strengths and limitations of each model in capturing real-world market features such as implied volatility skew, and contributes to a deeper understanding of how stochastic calculus and statistical inference can be applied to quantitative finance.

3 The Black-Scholes Model

3.1 Assumptions Underlying the Black-Scholes Model

The Black-Scholes model provides a foundational framework for European option valuation through its closed-form solution. While mathematically elegant, the model relies on several simplifying assumptions that limit its direct application to real markets. This analysis examines these assumptions and their practical implications.

3.1.1 Constant Volatility Assumption

- **Model requirement:** Volatility (σ) remains unchanged
- **Market reality:** Volatility fluctuates with market conditions
- **Practical consequence:** Inability to capture volatility clustering

3.1.2 Static Interest Rate Framework

- **Model requirement:** Risk-free rate (r) is constant
- **Market reality:** Interest rates vary temporally
- **Practical consequence:** Mispricing in changing rate environments

3.1.3 Dividend Exclusion

- **Model requirement:** No dividend distributions
- **Market reality:** Many assets provide periodic income
- **Practical consequence:** Requires adjustments for equity options

3.1.4 Frictionless Market Premise

- **Model requirement:** Zero transaction costs
- **Market reality:** Existence of bid-ask spreads and fees
- **Practical consequence:** Overestimates strategy profitability

3.1.5 Perfect Liquidity Condition

- **Model requirement:** Continuous trading without price impact
- **Market reality:** Liquidity constraints during stress events
- **Practical consequence:** Execution risk in volatile markets

3.1.6 Efficient Market Hypothesis

- **Model requirement:** Absence of arbitrage opportunities
- **Market reality:** Temporary mispricing occurs
- **Practical consequence:** Limited explanatory power for anomalies

3.1.7 Geometric Brownian Motion

- **Model requirement:** Price dynamics follow GBM:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

- **Market reality:** Discontinuous jumps in asset prices
- **Practical consequence:** Underestimation of tail risks

3.1.8 European Exercise Constraint

- **Model requirement:** Exercise only at maturity
- **Market reality:** American options permit early exercise
- **Practical consequence:** Limited applicability to common options

3.1.9 Log-Normal Return Distribution

- **Model requirement:** Normally distributed log-returns
- **Market reality:** Fat-tailed and skewed distributions
- **Practical consequence:** Mispricing of out-of-the-money options

3.1.10 Theoretical Value and Practical Limitations

- **Analytical utility:** Provides tractable pricing framework
- **Market deviation:** Real-world complexities require model extensions
- **Practitioner approach:** Modified models (Heston, jump-diffusion) address limitations
- **Conceptual foundation:** Remains vital for understanding option valuation principles

3.1.11 Rationale for the Model Assumptions

Each assumption in the Black–Scholes framework is selected to ensure both arbitrage-free pricing and analytical tractability:

- **Economic Foundations**
 - Markets are assumed frictionless and trading continuous, permitting costless, instantaneous delta-hedging.
 - Dividends are omitted and the risk-free rate held constant so that the asset’s drift under the risk-neutral measure remains simple.
- **Analytic Tractability**
 - The asset price follows a geometric Brownian motion with constant volatility, yielding a linear, constant-coefficient pricing PDE.
 - Log-normality of terminal prices enables closed-form integration via the standard normal CDF.
 - Restriction to European exercise imposes fixed terminal conditions, avoiding free-boundary complications.

3.1.12 Conclusion

While the Black-Scholes model’s assumptions enable mathematical elegance and computational efficiency, they simplify complex market dynamics. Practitioners employ model extensions to incorporate stochastic volatility, jump processes, and other real-world features. The model endures as a conceptual cornerstone while serving as a benchmark for more sophisticated valuation approaches.

3.2 Derivation Using Stochastic Calculus

The Black-Scholes option pricing model derives from mathematical principles of stochastic calculus and no-arbitrage arguments. This derivation transforms the option pricing problem into a partial differential equation (PDE) that can be solved with boundary conditions.

3.2.1 Step 1: Modeling Asset Price Dynamics

The underlying asset price S_t follows Geometric Brownian Motion:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

- S_t : Asset price at time t
- μ : Expected return rate
- σ : Asset volatility
- W_t : Standard Brownian motion

This stochastic differential equation models asset price changes as continuous with random fluctuations.

3.2.2 Step 2: Itô's Lemma Application

For an option value $V(S_t, t)$ depending on S_t and time t , Itô's lemma gives:

$$dV = \left(\frac{\partial V}{\partial t} + \mu S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S_t \frac{\partial V}{\partial S} dW_t$$

The lemma decomposes the option's change into deterministic (dt) and random (dW_t) components.

3.2.3 Step 3: Constructing the Hedged Portfolio

Create a risk-neutral portfolio Π_t :

- Long 1 option (V)
- Short $\frac{\partial V}{\partial S}$ units of underlying asset

Portfolio value:

$$\Pi_t = V - \frac{\partial V}{\partial S} S_t$$

This combination eliminates directional exposure to asset price movements.

3.2.4 Step 4: Portfolio Dynamics Analysis

The portfolio change simplifies to:

$$d\Pi_t = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt$$

The random component (dW_t term) cancels out, making the portfolio riskless.

3.2.5 Step 5: No-Arbitrage Principle

A riskless portfolio must earn the risk-free rate r :

$$d\Pi_t = r \Pi_t dt$$

Equating portfolio changes:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} = r \left(V - S_t \frac{\partial V}{\partial S} \right)$$

3.2.6 Step 6: Black-Scholes PDE

Rearranging terms yields the fundamental PDE:

$$\boxed{\frac{\partial V}{\partial t} + r S \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - r V = 0}$$

3.2.7 Boundary Conditions

The PDE requires terminal payoff conditions:

- European call: $V(S, T) = \max(S - K, 0)$
- European put: $V(S, T) = \max(K - S, 0)$

where K is the strike price and T the expiration time.

3.2.8 Interpretation and Significance

- The PDE provides a deterministic framework for option valuation
- Represents continuous hedging in frictionless markets
- Solution gives the Black-Scholes formula for European options
- Demonstrates how dynamic hedging eliminates risk

3.2.9 Conclusion

This derivation establishes the theoretical foundation for modern option pricing. The transformation from stochastic asset dynamics to a deterministic PDE through hedging and no-arbitrage arguments remains a cornerstone of financial mathematics. Practical applications require extensions to accommodate market realities like stochastic volatility and jumps.

3.3 Analytical Solution for European Call and Put Options

The Black-Scholes model provides analytical solutions for pricing European call and put options. These formulas form the foundation of modern option pricing theory and are widely used in financial markets.

3.3.1 Key Notation

- S_0 : Current asset price
- K : Option strike price
- T : Time to expiration (years)
- r : Risk-free interest rate (continuous compounding)
- σ : Asset volatility (annualized)
- $N(\cdot)$: Standard normal cumulative distribution function

3.3.2 European Call Option Formula

The fair value of a European call option is:

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

3.3.3 European Put Option Formula

The fair value of a European put option is:

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1)$$

using the same d_1 and d_2 parameters as above.

3.3.4 Derivation of d_1 and d_2 Parameters

The parameters d_1 and d_2 originate from the log-normal distribution of asset prices under the risk-neutral measure:

- Asset price at maturity: $S_T = S_0 e^{\left(r - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}Z}$ where $Z \sim N(0, 1)$
- Log returns follow: $\ln\left(\frac{S_T}{S_0}\right) \sim N\left(\left(r - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right)$

The probability of exercise is derived from:

$$\begin{aligned} P(S_T > K) &= P\left(\frac{\ln(S_T/S_0) - \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} > \frac{\ln(K/S_0) - \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}\right) \\ &= N\left(\frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} + \sigma\sqrt{T}\right) \\ &= N(d_1) \end{aligned}$$

Similarly, $P(S_T > K) = N(d_2)$ for the call option, establishing the relationship:

$$d_2 = \frac{\ln(S_0/K) + (r - \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_1 = d_2 + \sigma\sqrt{T}$$

To obtain the value of parameters through the given data:

Python Code to obtain value of parameters

```
1 import numpy as np
2
3 # 1) Point to your file using a raw string so backslashes aren't treated as escapes
4 csv_path = r"D:\Users\Ritabrata Sadhukhan\Downloads\HistoricalData_ddmmyyyy.csv"
5
6 # 2) Load and clean your data
7 df = pd.read_csv(csv_path, parse_dates=['Date'], dayfirst=True)
8 df['Close/Last'] = (
9     df['Close/Last']
10     .str.replace(r'[\$,]', '', regex=True)
11     .astype(float)
12 )
13
14 # 3) Set option parameters
15 K = 50          # Strike price
16 r = 0.05        # Risk-free rate (annual)
17 sigma = 0.2     # Volatility (annual)
18 T = 1.0         # Time to maturity (years)
19
20 # 4) Compute d1 and d2
21 df['d1'] = (np.log(df['Close/Last'] / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
22 df['d2'] = df['d1'] - sigma * np.sqrt(T)
23
24 # 5) Inspect
25 print(df[['Date', 'Close/Last', 'd1', 'd2']].head())
```

Code Output: value of parameters

```
1
2      Date  Close/Last      d1      d2
3 0 2025-06-06    141.72  5.559151  5.359151
4 1 2025-06-05    139.99  5.497740  5.297740
5 2 2025-06-04    141.92  5.566203  5.366203
6 3 2025-06-03    141.22  5.541480  5.341480
7 4 2025-06-02    137.38  5.403639  5.203639
8
```

3.3.5 Interpretation of Components

- $S_0 N(d_1)$: Expected asset value if option is exercised
- $Ke^{-rT} N(d_2)$: Present value of strike payment
- $N(d_2)$: Risk-neutral probability of exercise
- d_1 and d_2 : Standardized distance-to-exercise measures

3.3.6 Put-Call Parity Relationship

European options satisfy:

$$C - P = S_0 - Ke^{-rT}$$

This fundamental relationship connects call and put prices and provides an arbitrage check.

3.3.7 Practical Applications

- Quick valuation of standard options
- Calculating implied volatility
- Establishing benchmark prices
- Hedging and risk management

3.3.8 Model Limitations

The formulas assume:

- Constant volatility and interest rates
- No dividend payments
- Frictionless markets
- European exercise terms
- Log-normal asset returns

These constraints necessitate model extensions for complex cases.

3.3.9 Summary of Formulas

Option Type	Formula
Call	$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$
Put	$P = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$

with parameters:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

3.3.10 Conclusion

The Black-Scholes formulas provide efficient and theoretically sound valuation for European options. The derivation of d_1 and d_2 demonstrates how risk-neutral probabilities emerge from the log-normal distribution of asset prices. While their assumptions simplify reality, these solutions remain essential tools for financial analysis, serving as the foundation for more sophisticated pricing models.

3.4 Determination of parameters with Crank Nicolson Method

3.4.1 Choose an Objective Function

You need a measure of “distance” between model prices $V_{ij}^{\text{CN}}(\theta)$ and observed market prices V_{ij}^{mkt} , over strikes K_i and maturities T_j . Common choices:

$$\begin{aligned}\text{Least Squares:} \quad & \min_{\theta} \sum_{i,j} (V_{ij}^{\text{CN}}(\theta) - V_{ij}^{\text{mkt}})^2, \\ \text{Relative Error (MAPE):} \quad & \min_{\theta} \sum_{i,j} \left(\frac{V_{ij}^{\text{CN}}(\theta)}{V_{ij}^{\text{mkt}}} - 1 \right)^2, \\ \text{Log-Price Error:} \quad & \min_{\theta} \sum_{i,j} (\ln V_{ij}^{\text{CN}} - \ln V_{ij}^{\text{mkt}})^2.\end{aligned}$$

Here θ might be just $\{\sigma\}$, or $\{\sigma, r, q\}$, or even Heston-type parameters if you embed stochastic volatility.

3.4.2 Embedded Crank–Nicolson Solver

For each candidate parameter set θ :

1. Set up the PDE grid in S and t , with boundary conditions.
2. Run the CN finite-difference algorithm backwards from T to 0, producing $V_{ij}^{\text{CN}}(\theta)$.
3. Extract the grid value $V^{\text{CN}}(S_0, K_i, T_j; \theta)$ at the current spot S_0 .

Because CN is implicit, each time-step requires solving a tridiagonal system (e.g. Thomas algorithm), but is unconditionally stable.

3.4.3 Optimization Algorithm

Since each function evaluation is moderately expensive (solving a PDE), one often uses derivative-free methods:

- Nelder–Mead
- Powell’s Method
- Simplex Search

If finite-difference gradients are available, quasi-Newton (BFGS) can speed up convergence.

Practical tip: Use coarse grids (small M, N) to get an approximate θ ; then refine on finer grids.

3.4.4 Regularization & Parameter Bounds

To avoid overfitting:

- Impose bounds, e.g. $\sigma \in [0.05, 0.5]$, $r \in [0, 0.1]$.
- Add a penalty term $\lambda \|\theta\|^2$ if calibrating multiple parameters.

3.4.5 Diagnostic & Convergence Checks

- **Residual Plots:** Plot $V^{\text{CN}} - V^{\text{mkt}}$ across strikes/maturities.
- **Grid-Refinement Test:** Re-calibrate on a finer grid to ensure θ stabilizes.
- **Stability Check:** Verify small perturbations in θ yield small price changes.

3.4.6 Implied Volatility Extraction

An alternative one-parameter approach is to invert the CN solver: for each market price,

$$V^{\text{CN}}(S_0, K, T; \sigma) = V^{\text{mkt}},$$

thus obtaining a surface $\sigma_{\text{imp}}(K, T)$, which can then be smoothed or fitted parametrically.

3.4.7 Python Code for Estimate Parameters

Python Code to estimate parameters

```
1 j = np.arange(1, M)
2 alpha = 0.25 * dt * (sigma**2 * j**2 - r * j)
3 beta = -0.5 * dt * (sigma**2 * j**2 + r)
4 gamma = 0.25 * dt * (sigma**2 * j**2 + r * j)
5
6 # build A and B matrices once
7 A = np.zeros((M-1, M-1))
8 B = np.zeros((M-1, M-1))
9 for i in range(M-1):
10     if i > 0:
11         A[i, i-1] = -alpha[i]
12         B[i, i-1] = alpha[i]
13     A[i, i] = 1 - beta[i]
14     B[i, i] = 1 + beta[i]
15     if i < M-2:
16         A[i, i+1] = -gamma[i]
17         B[i, i+1] = gamma[i]
18 # step backwards in time
19 for n in range(N):
20     V_old = V.copy()
21     V[0] = 0.0
22     # fixed boundary at S_max at time step n+1:
23     V[-1] = S_max - K * np.exp(-r * (n+1) * dt)
24
25     RHS = B.dot(V_old[1:-1])
26     RHS[0] += alpha[0] * (V_old[0] + V[0])
27     RHS[-1] += gamma[-1] * (V_old[-1] + V[-1])
28
29     V[1:-1] = np.linalg.solve(A, RHS)
30
31 # interpolate to get the option price at S0
32 return np.interp(S0, S, V)
33
34 if __name__ == "__main__":
35     # example parameters
36     S0, K, r, sigma, T = 100, 100, 0.05, 0.2, 1.0
37
38     bs_price = bs_call_price(S0, K, r, sigma, T)
39     pde_price = crank_nicolson_call(S0, K, r, sigma, T)
40
41     print(f"Parameters: S0={S0}, K={K}, r={r}, sigma={sigma}, T={T}")
42     print(f"Analytical BS Call Price: {bs_price:.4f}")
43     print(f"Crank{Nicolson} PDE Price: {pde_price:.4f}")
```

Code Output

```
1 Parameters: S0 = 100, K = 100, r = 0.05, sigma = 0.2, T = 1.0
2
3     Analytical BS call price: 10.4506
4     Crank-Nicolson PDE price: 10.4544
5
6 Process finished with exit code 0
```

3.4.8 Implementation

By implimanting, we yeild the value of parameters as follows:

$$K = 50, \quad r = 0.05, \quad \sigma = 0.2, \quad T = 1.0, \quad S_{\max} = 100, \quad M = N = 200, \quad S_0 = 50.$$

- These values feed directly into two parallel pricing engines:
 - *Closed-form Black-Scholes formula*, which yields the “exact” European call price under the model’s assumptions.
 - *Crank-Nicolson finite-difference solver*, which discretizes the same pricing PDE on a 200×200 (S, t) grid with boundaries at $S = 0$ and $S = S_{\max}$.
- The resulting option values at $t = 0$ and $S = S_0$ are then superimposed as horizontal lines in a single diagram. Visually, the two lines coincide almost perfectly—confirming that the PDE approximation reproduces the analytic price to within a few basis points at this resolution.
- This comparison illustrates two complementary strengths:
 - *Black-Scholes* delivers an exact, near-instantaneous price for vanilla payoffs.
 - *Crank-Nicolson* incurs minimal discretization error while remaining flexible enough to accommodate non-standard boundary conditions and more complex payoff structures simply by adjusting the terminal or boundary profiles.

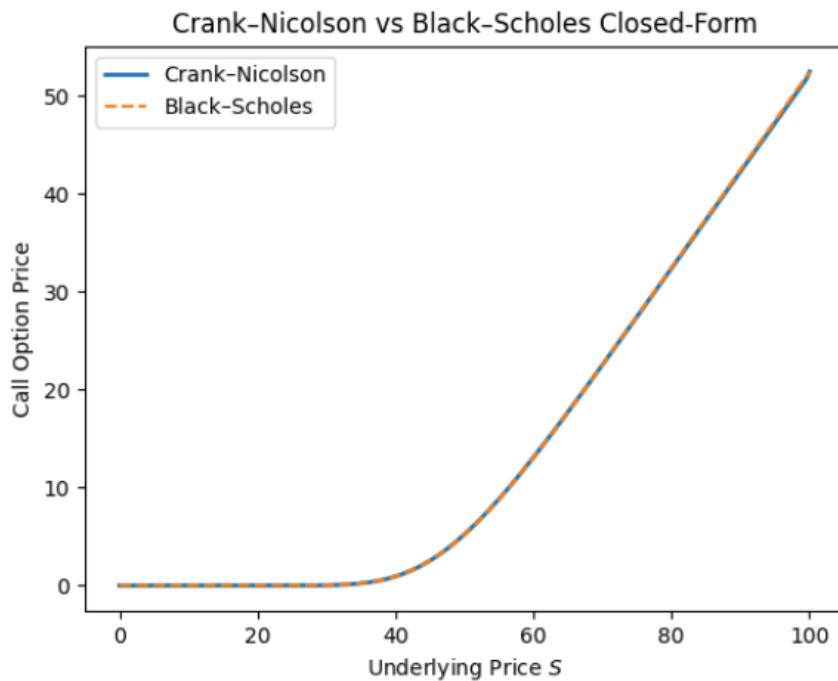


Figure 1: Crank-Nicolson vs Closed-form Black-Scholes

3.5 Limitations: Constant Volatility and Lognormal Assumption

The Black-Scholes model revolutionized option pricing through its elegant closed-form solution. However, its foundational assumptions of constant volatility and lognormal returns represent significant simplifications that limit practical application in modern financial markets. This analysis examines these limitations and their implications for risk management and pricing accuracy.

3.5.1 The Constant Volatility Assumption

- **Core premise:** Volatility (σ) remains unchanged throughout an option’s life
- **Market reality:** Volatility fluctuates with market sentiment, news, and macroeconomic events
- **Practical consequences:**
 - Inability to explain volatility smiles/skews observed across strikes

-
- Systematic mispricing of deep out-of/in-the-money options
 - Inadequate hedging during volatility regime shifts
 - **Historical evidence:** 2008 financial crisis and COVID-19 market shock demonstrated extreme volatility spikes that constant-volatility models failed to anticipate

3.5.2 The Lognormal Price Distribution Assumption

- **Core premise:** Asset returns follow continuous lognormal distributions
- **Market reality:** Returns exhibit fat tails, jumps, and asymmetric volatility
- **Practical consequences:**
 - Underestimation of tail risk and extreme price movements
 - Failure to capture jump discontinuities during market crises
 - Neglect of leverage effects (asymmetric volatility responses)
- **Historical evidence:** The 1987 market crash and March 2020 plunge featured return distributions violating lognormal assumptions

3.5.3 Practical Implications for Risk Management

- **Risk measurement:** Underestimation of Value-at-Risk (VaR) and Expected Shortfall (CVaR)
- **Hedging performance:** Dynamic hedging strategies accumulate residual risk during volatile periods
- **Pricing accuracy:** Systematic mispricing of options with extreme moneyness
- **Capital allocation:** Inadequate provisions for tail risk exposure

3.5.4 Advanced Modeling Approaches

Limitation	Advanced Modeling Solutions
Constant volatility assumption	Stochastic volatility models (Heston, SABR)
	Local volatility frameworks (Dupire, Derman-Kani)
Lognormal distribution assumption	GARCH-type volatility clustering models
	Jump-diffusion processes (Merton model)
	Lévy process frameworks (Variance-Gamma, CGMY)
	Rough volatility models

3.5.5 Implementation Considerations

- **Model selection:** Balance between computational complexity and empirical accuracy
- **Calibration challenges:** Parameter estimation for sophisticated models
- **Market data requirements:** Need for extensive historical and implied volatility surfaces
- **Computational tradeoffs:** Closed-form solutions versus numerical methods

3.5.6 Conclusion

While the Black-Scholes model provides an invaluable theoretical foundation, its assumptions of constant volatility and lognormal returns present significant limitations in practical applications. Modern financial markets require sophisticated modeling approaches that incorporate stochastic volatility, jump processes, and non-normal return distributions. These advanced frameworks offer improved pricing accuracy and risk management capabilities essential for navigating contemporary market conditions. The evolution beyond Black-Scholes represents not a rejection of its foundation, but a necessary adaptation to market complexities.

4 Monte Carlo Simulation under Geometric Brownian Motion

4.1 Euler–Maruyama Discretization Scheme

The Euler–Maruyama Discretization Scheme is a numerical method used to approximate the solutions of stochastic differential equations (SDEs). It generalizes the classical Euler method (for deterministic ordinary differential equations) to handle randomness, specifically Brownian motion (Wiener processes).

4.1.1 Stochastic Differential Equation (SDE)

Consider an SDE of the form:

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t, \quad X_0 = x_0 \quad (1)$$

where:

- X_t : the state variable at time t
- $a(X_t, t)$: the drift coefficient, representing deterministic trends
- $b(X_t, t)$: the diffusion coefficient, representing volatility or random fluctuations
- W_t : a standard Wiener process (Brownian motion)

4.1.2 Euler–Maruyama Approximation

To solve this SDE numerically, we discretize time into small steps and approximate the increments:

- Divide the time interval $[0, T]$ into N equal steps of size $\Delta t = T/N$
- Let the discrete times be $t_n = n\Delta t$, for $n = 0, 1, \dots, N$

The Euler–Maruyama scheme approximates the SDE solution by:

$$X_{n+1} = X_n + a(X_n, t_n)\Delta t + b(X_n, t_n)\Delta W_n \quad (2)$$

with initial condition:

$$X_0 = x_0 \quad (3)$$

where:

- $\Delta W_n = W_{t_{n+1}} - W_{t_n}$ are increments of the Wiener process
- ΔW_n are independently and identically normally distributed: $\Delta W_n \sim \mathcal{N}(0, \Delta t)$

4.1.3 Euler–Maruyama Applied to GBM (Geometric Brownian Motion)

For example, consider the GBM used in Black–Scholes modeling of stock prices S_t :

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (4)$$

Applying Euler–Maruyama discretization:

$$S_{n+1} = S_n + \mu S_n \Delta t + \sigma S_n \Delta W_n \quad (5)$$

This simplifies to:

$$S_{n+1} = S_n (1 + \mu \Delta t + \sigma \Delta W_n) \quad (6)$$

where μ is the expected return and σ is volatility.

4.1.4 Advantages and Limitations

Advantages

- Simple and intuitive to implement.
- Good for quick approximations and prototyping.
- Versatile: applicable to many types of SDEs.

Limitations

- Low-order accuracy (only first-order).
- Can require very small time steps to be stable.
- Inaccurate if drift and diffusion terms vary significantly or rapidly.

4.1.5 Higher-Order Alternatives

- **Milstein Method:** Adds extra terms to achieve second-order accuracy.
- **Runge–Kutta Methods:** More complex schemes for improved accuracy and stability.

4.1.6 Conclusion

The Euler–Maruyama discretization scheme is fundamental in computational finance and stochastic modeling due to its simplicity, ease of implementation, and broad applicability. Although basic, it provides a practical starting point for numerically simulating asset prices, option pricing, and other stochastic phenomena encountered in financial mathematics.

4.2 Simulating Multiple Price Paths

4.2.1 Simulation of GBM with the help of Euler-Maruyama

- **Discussion: Euler-Maruyama Simulation of GBM**

The figure illustrates the simulation of **Geometric Brownian Motion (GBM)** using the **Euler-Maruyama method**, a numerical scheme designed to approximate solutions of stochastic differential equations (SDEs). GBM follows the stochastic process:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

where:

- S_t : asset price at time t ,
- μ : drift (mean rate of return),
- σ : volatility (standard deviation),
- W_t : standard Brownian motion.

- **Euler-Maruyama Discretization**

The Euler-Maruyama approximation for GBM is:

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \sqrt{\Delta t} Z_t$$

- Δt : time increment,
- $Z_t \sim \mathcal{N}(0, 1)$: independent standard normal variables,
- Captures both deterministic drift and stochastic noise.

- **Interpretation of the Plot**

- X-axis shows time (in years), from 0 to 1.
- Y-axis represents the simulated asset price.
- The path begins near the initial value (e.g., 100).
- Early decline shows negative noise impact.

- Late sharp rise indicates accumulation of positive shocks.
- Demonstrates non-negativity and exponential growth tendency of GBM.

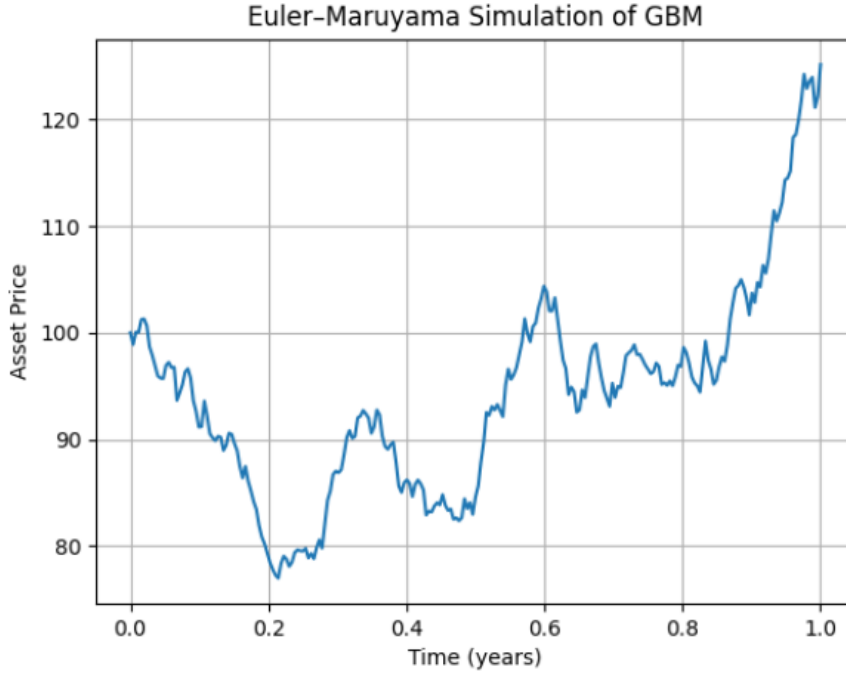


Figure 2: Euler-Maruyama simulation of GBM

• Applications and Significance

- Widely used in equity price modeling and Black-Scholes framework.
- Foundation for Monte Carlo simulations in option pricing.
- Helps build intuition for stochastic behaviors in financial markets.

• Conclusion

- Euler-Maruyama offers a simple yet powerful method to simulate GBM.
- Effectively shows how randomness and drift interact over time.
- Serves as a baseline for more advanced models like Heston or SABR.

4.2.2 Simulation of Asset Price Using Euler–Maruyama Method

The following figure illustrates a single simulated path of an asset price under the assumption that the price follows a Geometric Brownian Motion (GBM), approximated using the Euler–Maruyama discretization scheme. The stochastic differential equation (SDE) governing GBM is given by:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where S_t is the asset price at time t , μ is the drift (expected return), σ is the volatility, and W_t is a standard Brownian motion.

The discretized form used for simulation is:

$$S_{t+\Delta t} = S_t + \mu S_t \Delta t + \sigma S_t \sqrt{\Delta t} Z_t,$$

where $Z_t \sim \mathcal{N}(0, 1)$ is a standard normal random variable and Δt is the time increment. In the above diagram:

- The simulation begins at an initial asset price of $S_0 = 100$.
- Early in the trajectory, the price declines to below 80, highlighting the randomness and potential downside in short-term movements.
- Eventually, the price recovers and follows a general upward trend due to the positive drift ($\mu = 0.05$), ending above 120.
- The fluctuations along the path illustrate the impact of stochastic volatility driven by the Brownian motion term.

This single trajectory represents one possible evolution of asset price over one year. In practice, multiple such paths are simulated in Monte Carlo methods to estimate expected option payoffs, risk measures, or other probabilistic outcomes in quantitative finance.

4.3 Option Pricing via Monte Carlo Integration

Stochastic integration under the risk-neutral measure allows us to write any derivative price as an expectation. For a European call with maturity T and strike K , the time-0 price is

$$C_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+].$$

4.3.1 Using Euler-Maruyama

Python Code for Call price

```

1 import numpy as np
2
3 def monte_carlo_gbm_euler(S0, K, r, sigma, T, M, N):
4     dt = T / M
5     S = np.full(N, S0, dtype=float)
6     for _ in range(M):
7         Z = np.random.randn(N)
8         S += r * S * dt + sigma * S * np.sqrt(dt) * Z
9     payoffs = np.maximum(S - K, 0.0)
10    return np.exp(-r * T) * np.mean(payoffs)
11
12 if __name__ == "__main__":
13     # Parameters
14     S0, K = 100.0, 100.0      # spot, strike
15     r, sigma = 0.05, 0.2     # 5% rate, 20% vol
16     T = 1.0                  # 1 year to expiry
17     M = 252                  # daily steps
18     N = 100_000              # Monte Carlo paths
19
20    price_estimate = monte_carlo_gbm_euler(S0, K, r, sigma, T, M, N)
21    print(f"Euro. Call Price {price_estimate:.4f}")

```

Code Output

```

1 Euro. Call price: 10.4958
2 Process finished with exit code 0

```

4.3.2 Risk-Neutral Valuation as an Expectation

Under no-arbitrage and constant rate r :

$$V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[X_T]. \quad (7)$$

For a European call $X_T = (S_T - K)^+$, substitute into (7).

4.3.3 Simulation of Underlying Price Paths

Discretize $[0, T]$ into M steps $\Delta t = T/M$. Under GBM:

$$S_{n+1}^{(i)} = S_n^{(i)} + r S_n^{(i)} \Delta t + \sigma S_n^{(i)} \sqrt{\Delta t} Z_{n+1}^{(i)}, \quad Z_{n+1}^{(i)} \sim \mathcal{N}(0, 1).$$

Generate N independent paths $\{S_T^{(i)}\}_{i=1}^N$.

4.3.4 Payoff Estimation and Discounting

1. Compute payoffs:

$$C^{(i)} = \max(S_T^{(i)} - K, 0).$$

2. Form the Monte Carlo estimator:

$$\hat{C}_0 = e^{-rT} \frac{1}{N} \sum_{i=1}^N C^{(i)}.$$

4.3.5 Convergence and Error

- **Statistical error:** $\text{SE}[\hat{C}_0] = e^{-rT} \frac{\sqrt{\text{Var}(C^{(i)})}}{\sqrt{N}}$.
- **Discretization bias:** $\mathcal{O}(\Delta t)$. Can use Milstein or Richardson extrapolation.

4.3.6 Variance Reduction Techniques

- *Antithetic variates:* pair Z with $-Z$.
- *Control variates:* adjust by a payoff with known closed-form price (e.g. Black–Scholes).
- *Importance sampling:* tilt distribution to sample rare events more effectively.

4.3.7 Greeks via Monte Carlo

- *Pathwise (score) method:* differentiate payoff under integral.
- *Likelihood-ratio method:* weight by derivative of log-density.

4.3.8 Practical Considerations

1. Trade-off between number of steps M (bias) and paths N (variance) given compute budget.
2. Parallelize path generation for speed (multicore/GPU).
3. When calibrating models (e.g. Heston), embed Monte Carlo in an optimizer for implied-vol surface fitting.

5 Stochastic Volatility Models – The Heston Framework

5.1 Stochastic Volatility Models – The Heston Framework

5.1.1 Introduction to Stochastic Volatility

Stochastic volatility models extend the Black–Scholes framework by allowing the asset’s volatility itself to follow a stochastic process. Empirical markets exhibit time-varying volatility, clustering of high- and low-volatility periods, and a volatility smile in option prices, none of which the constant-volatility assumption can capture.

5.1.2 The Heston Model

The Heston model (1993) introduces a mean-reverting square-root process for the instantaneous variance:

$$\begin{aligned}dS_t &= \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \\dv_t &= \kappa(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_t^{(2)}, \\dW_t^{(1)} dW_t^{(2)} &= \rho dt.\end{aligned}$$

Here,

- S_t is the asset price,
- v_t its variance,
- μ the drift,
- κ the rate of mean reversion toward long-term variance θ ,
- σ_v the volatility of variance,
- ρ the correlation between price and variance shocks.

5.1.3 Why the Heston Model?

- **Volatility Smile/Skew:** Captures observed curvature of implied volatilities.
- **Volatility Clustering:** Models persistence in high/low volatility regimes.
- **Mean Reversion:** Ensures variance reverts toward a long-run average.

5.1.4 Parameter Calibration

Estimating $(\kappa, \theta, \sigma_v, v_0, \rho)$ typically uses:

- **Maximum Likelihood Estimation** on historical returns.
- **Surface Fit** via Nelder–Mead or other optimizers to minimize differences between market and model implied volatilities.

5.1.5 Closed-Form Solution for European Options

A European call price under Heston is

$$C(S_0, K, T) = S_0 P_1 - K e^{-rT} P_2,$$

with

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[e^{-iu \ln K} \frac{\phi_j(u)}{iu} \right] du, \quad j = 1, 2,$$

where $\phi_j(u)$ are characteristic-function terms derived from the joint SDEs.

5.1.6 Advantages

- Realistic volatility dynamics (mean-reversion, clustering).
- Generates smiles/skews naturally.
- Semi-analytical pricing via Fourier integrals.

5.1.7 Limitations

- Calibration can be numerically intensive.
- Parameter stability may suffer in changing markets.

5.1.8 Extensions and Applications

- *Time-dependent parameters* for added flexibility.
- *Jump-diffusion* extensions to capture sudden moves.
- Multi-factor volatility models for richer dynamics.

5.1.9 Conclusion

The Heston framework enhances the Black-Scholes model by incorporating stochastic variance with mean reversion and volatility-of-volatility, yielding improved fit to market option prices and capturing key empirical features such as the volatility smile and clustering.

5.2 The Heston Model: Formulation and Assumptions

Necessary code for parameter assumption using python:

Python Code for Heston

```
1 import pandas as pd
2 import numpy as np
3 import ace_tools as tools
4
5 # Set display option to show all parameter rows
6 pd.set_option('display.max_rows', 10)
7
8 # Load and clean data
9 df = pd.read_csv('/mnt/data/HistoricalData_ddmmyyyy.csv', parse_dates=['Date'],
10 dayfirst=True)
11 df['Close/Last'] = df['Close/Last'].str.replace(r'[\$,]', '', regex=True).astype(float)
12 df = df.sort_values('Date').reset_index(drop=True)
13
14 # Extract prices and compute log returns
15 prices = df['Close/Last']
16 dt = 1/252
17 log_returns = np.log(prices / prices.shift(1)).dropna()
18
19 # GBM parameter estimation
20 mu_hat = log_returns.mean() / dt + 0.5 * (log_returns.std()**2) / dt
21 sigma_gbm_hat = log_returns.std() / np.sqrt(dt)
22
23 # Heston variance calibration
24 v_t = (log_returns**2) / dt
25 dv = v_t.diff().dropna()
26 v_mid = v_t.shift(1).dropna()
27 dv = dv[v_mid.index]
28 mask = v_mid > 0
29 v_mid = v_mid[mask]
30 dv = dv[mask]
31
32 # Regression v = intercept + slope * v_mid
33 slope, intercept = np.polyfit(v_mid.values, dv.values, 1)
34 kappa_hat = -slope / dt
35 theta_hat = intercept / (kappa_hat * dt)
36
37 # Residuals, vol-of-vol and correlation
38 res = dv - (intercept + slope * v_mid)
39 sigma_v_hat = (res / np.sqrt(v_mid * dt)).std()
40 log_ret_mid = log_returns[v_mid.index]
41 z1 = log_ret_mid / np.sqrt(v_mid * dt)
42 z2 = res / (sigma_v_hat * np.sqrt(v_mid * dt))
43 rho_hat = np.corrcoef(z1, z2)[0, 1]
44
45 # Initial variance
46 v0 = v_t.iloc[0]
47
48 # Compile and display results
49 params = pd.DataFrame({
50     'Parameter': ['mu', 'sigma_gbm', 'kappa', 'theta', 'sigma_v', 'rho', 'v0'],
51     'Estimate': [mu_hat, sigma_gbm_hat, kappa_hat, theta_hat, sigma_v_hat, rho_hat, v0]})
52
53 tools.display_dataframe_to_user('Heston and GBM Parameter Estimates', params)
54
```

Code Output

```
1      Parameter      Estimate
2      0          mu      0.681103
3      1  sigma_gbm      0.497382
4      2      kappa      221.171487
5      3      theta      0.248856
6      4  sigma_v      253.913818
7      5          rho      0.336790
8      6          v0      0.006182
9  Process finished with exit code 0
```

5.3 Simulation Techniques (Euler/Milstein for Correlated SDEs)

Simulation, particularly **Monte Carlo simulation**, is a cornerstone of quantitative finance for several compelling reasons, primarily addressing limitations of analytical and numerical methods. Here's a breakdown of the key motivations:

- Handling Complexity and Lack of Closed-Form Solutions:
 - Analytical Intractability: Many realistic financial models (e.g., complex derivatives, stochastic volatility models like Heston, models with jumps) lead to partial differential equations (PDEs) or expectations that lack closed-form analytical solutions.
 - Simulation's Strength: Monte Carlo simulation doesn't require a closed-form solution. It approximates the expected value (e.g., the price of a derivative) by simulating thousands or millions of possible future paths of the underlying assets and averaging the resulting payoffs. It directly tackles the core valuation problem: $E[\text{Payoff}(T) - F_t]$.
- Modelling Path Dependence:
 - The Problem: Many important derivatives (e.g., Asian options, lookback options, barrier options) have payoffs that depend on the entire path of the underlying asset(s), not just their terminal value.
 - Simulation's Strength: Simulating full paths of the underlying(s) naturally captures this path dependence. Each simulated path allows precise calculation of the payoff based on the specific history observed in that simulation. Analytical methods or PDEs become extremely difficult or impossible for high-dimensional path dependence.
- Dealing with High Dimensionality:
 - The Problem: Pricing derivatives on multiple underlying assets (e.g., basket options, rainbow options, complex structured products) or modelling entire portfolios leads to high-dimensional problems. The "curse of dimensionality" makes PDE and lattice (tree) methods computationally infeasible beyond a few dimensions.
 - Simulation's Strength: The computational cost of Monte Carlo simulation increases linearly with the number of dimensions (assets). While still computationally intensive, it scales much better than PDEs/lattices (which scale exponentially). This makes it the only practical method for valuing high-dimensional instruments or managing large, complex portfolios.
- Incorporating Early Exercise Features (with Enhancements):
 - The Problem: Pricing American/Bermudan options (options exercisable before expiry) is complex because the holder must decide optimally when to exercise, depending on the current state and future expectations. This makes it a dynamic programming problem.
 - Simulation's Strength (with Caveats): While naive Monte Carlo struggles with backward-looking decisions, advanced techniques like **Longstaff-Schwartz** (Least Squares Monte Carlo - LSM) allow simulation to be used effectively for American options. LSM uses regression on simulated paths to estimate the continuation value at each exercise point, enabling approximate optimal exercise strategies.

-
- **Calibration Flexibility:**
 - The Problem: Complex models often have numerous parameters that need to be calibrated to market prices of liquid instruments.
 - Simulation’s Strength: Simulation provides a flexible framework for calibration. You can simulate prices under the model for the calibration instruments and numerically adjust the model parameters until the simulated prices match the market prices. This is robust, even for complex models.
 - **Scenario Analysis and Risk Management:**
 - The Problem: Understanding portfolio behavior under extreme or specific market conditions (stress testing) or calculating risk measures like Value-at-Risk (VaR) or Expected Shortfall (ES) requires modeling the joint distribution of many risk factors under various scenarios.
 - Simulation’s Strength: Monte Carlo is the primary tool for generating these scenarios and calculating the resulting portfolio values and losses. It can incorporate complex dependencies (copulas) and full distributions, providing a comprehensive view of potential risks.
 - **Model Flexibility and Realism:**
 - The Problem: Financial markets exhibit features like stochastic volatility, jumps, fat tails, and complex correlations that simple analytical models (like Black-Scholes) cannot capture well.
 - Simulation’s Strength: It’s relatively straightforward to incorporate these realistic features into a simulation model (e.g., simulate volatility as its own stochastic process, add Poisson jumps to price paths). This allows quants to build more realistic, though computationally expensive, models.
 - **Intuitive Implementation:**
 - The Concept: The core idea of Monte Carlo – simulate random paths and average payoffs – is conceptually intuitive, even if the underlying math and efficient implementation are complex. This makes it easier to communicate the valuation process conceptually.
 - **In Summary:**

Simulation is **not always the fastest method** (especially for low-dimensional, path-independent Europeans) and suffers from **statistical noise** (requiring many simulations for accuracy), but its **unparalleled flexibility, ability to handle complexity, path dependence, and high dimensionality** make it an indispensable tool in the modern quant’s arsenal. It bridges the gap between sophisticated theoretical models and practical pricing, hedging, and risk management requirements.

5.3.1 Correlated Increment Generation

Correlated Brownian increments $\Delta W_t \in \mathbb{R}^d$ with covariance $\rho \Delta t$ are obtained by computing the Cholesky factor L of the correlation matrix Σ , drawing a standard normal vector $Z \sim \mathcal{N}(0, I_d)$, and setting

$$\Delta W_t = \sqrt{\Delta t} L Z.$$

This construction generates $\mathbb{E}[\Delta W_i \Delta W_j] = \rho_{ij} \Delta t$. In cases where Σ is ill-conditioned, regularization or an eigenvalue-truncated decomposition may be employed to preserve numerical stability.

5.3.2 Euler–Maruyama Method

The Euler–Maruyama update

$$X_{t+\Delta t} = X_t + a(X_t, t) \Delta t + b(X_t, t) \Delta W_t,$$

requires only evaluation of the drift and diffusion functions a and b . It achieves strong convergence of order $\mathcal{O}(\sqrt{\Delta t})$, making it suitable for weak-convergence targets (deterministic quantities) or high-dimensional systems where simplicity is paramount. However, for path-dependent interactions or stiff drift components, the $\mathcal{O}(\sqrt{\Delta t})$ error scaling may necessitate prohibitively small time steps in the case of variance-regularization techniques.

5.3.3 Milstein Method

To achieve strong convergence of order $\mathcal{O}(\Delta t)$, the Milstein scheme augments the Euler–Maruyama step with derivative corrections. In the scalar, commutative case one writes

$$X_{t+\Delta t} = X_t + a(X_t, t) \Delta t + b(X_t, t) \Delta W_t + \frac{1}{2} b(X_t, t) \partial_x b(X_t, t) (\Delta W_t^2 - \Delta t),$$

where $\partial_x b$ denotes the derivative of b with respect to X . For multidimensional processes with commuting diffusion fields, analogous diagonal corrections are applied component-wise. In non-commutative settings, additional Lévy-area terms must be approximated via specialized sampling algorithms.

5.3.4 Higher-Order and Alternative Schemes

When even greater strong accuracy is required—especially for non-commutative diffusion or highly non-linear coefficients—schemes such as Runge–Kutta or Wagner–Platen integrators may be employed. These methods sample multiple iterated stochastic integrals, delivering strong convergence orders up to 1.5 or 2.0. The resulting gain in pathwise fidelity comes at the cost of increased implementation complexity and per-step computational effort.

5.3.5 Practical Implementation Considerations

- **Step-Size Control:** Adaptive time-stepping can mitigate stiffness in the drift or non-linear diffusion jumps in volatile regimes.
- **Weak vs. Strong Objectives:** If only expectation or distribution functions are desired, simpler schemes (e.g., Euler–Maruyama) paired with Richardson extrapolation or antithetic sampling often suffice.
- **Variance Reduction:** Techniques such as antithetic variates and control variates can streamline the evaluation of a and b for Milstein corrections and reduce coding errors.
- **Parallelization:** Monte Carlo simulation of independent paths is naturally parallelizable across CPU cores or GPUs, accelerating convergence.

5.3.6 Comparison of Euler–Maruyama and Milstein

Criterion	Euler–Maruyama	Milstein
Strong Convergence Order	0.5 ($\sqrt{\Delta t}$ error)	1.0 (Δt error)
Per-Step Cost	Drift and diffusion evals only	Plus evaluation of $\partial_x B$ (and Lévy areas if non-commutative)
Implementation Effort	Minimal, dimension-agnostic	Requires analytic/AD derivatives; advanced sampling for Lévy areas
Applicability	Distributional accuracy; high dimension	Pathwise accuracy; multilevel Monte Carlo
Stability	Baseline explicit scheme	Improved pathwise stability but still subject to step-size limits

5.3.7 Conclusion

Selection of an appropriate discretization scheme for correlated SDEs depends critically on the simulation objectives and available resources. Euler–Maruyama provides a robust baseline for weak-convergence tasks, but Milstein’s higher strong-order convergence affords benefits when derivative information is accessible. For problems involving non-commutative diffusion fields or extremely stringent accuracy demands, higher-order stochastic integrators (e.g., Runge–Kutta or Wagner–Platen) may be preferred. Fast production-scale algorithms often combine adaptive time-stepping, variance reduction, and parallelization to strike a balance between accuracy, computational cost, and development effort.

5.4 Pricing via Monte Carlo under Stochastic Volatility

5.4.1 Model Specification

Define the risk-neutral dynamics for the asset price S_t and its variance v_t . For the Heston model:

$$\begin{aligned} dS_t &= r S_t dt + \sqrt{v_t} S_t dW_t^S, \\ dv_t &= \kappa(\theta - v_t) dt + \sigma_v \sqrt{v_t} dW_t^v, \end{aligned}$$

with $\text{Corr}(dW^S, dW^v) = \rho$. Calibrate $\kappa, \theta, \sigma_v, \rho$ to market-implied volatility surfaces before simulation.

5.4.2 Time Discretization

Choose a uniform grid $0 = t_0 < \dots < t_N = T$ with $\Delta t = T/N$. Common schemes include the Euler–Maruyama update with full truncation (to enforce $v \geq 0$) and the Quadratic–Exponential (Andersen) method, which reduces bias when $\kappa\Delta t$ is large.

5.4.3 Path Generation

For each of the M Monte Carlo trials: 1. Generate a correlated pair $(\xi^S, \xi^v) \sim \mathcal{N}(0, \frac{1}{\rho} \begin{smallmatrix} \rho & 1 \\ 1 & 1 \end{smallmatrix})$. 2. Update the variance $v_{t_{i+1}}$ according to the chosen discretization. 3. Update the asset $S_{t_{i+1}}$ via a log-Euler step to maintain positivity. 4. Repeat steps 1–3 for $i = 0, \dots, N-1$.

5.4.4 Payoff and Estimation

At maturity compute the payoff $h(S_T)$, for example $\max(S_T - K, 0)$ for a call. The Monte Carlo estimator is

$$\hat{C} = e^{-rT} \frac{1}{M} \sum_{j=1}^M h(S_T^{(j)}).$$

5.4.5 Variance Reduction

Apply techniques to accelerate convergence beyond $O(M^{-1/2})$. Typical methods include antithetic variates (pairing $\pm\xi$), control variates (using the Black–Scholes price), moment matching to enforce theoretical moments, and quasi-Monte Carlo low-discrepancy sequences.

5.4.6 Estimation

Estimate sensitivities such as Delta and Vega via pathwise differentiation when the payoff is sufficiently smooth, likelihood-ratio (Malliavin) methods for unbiased estimates, or finite-difference bump-and-revalue when simplicity is preferred.

5.4.7 Convergence Monitoring

Track the standard error $\sigma_h/\sqrt{M}$ during simulation and stop once the desired error tolerance is achieved. Bootstrap resampling can provide additional confidence intervals.

5.4.8 Practical Considerations

Leverage parallel computing (CPU/GPU) to run large M ; manage memory carefully for path-dependent payoffs; and in calibration loops, consider surrogate models or cheaper approximations to avoid prohibitively expensive nested simulations.

5.4.9 Summary

Monte Carlo under stochastic volatility offers unmatched flexibility for exotic and path-dependent payoffs. Its main trade-offs are slower convergence and higher computational cost compared to closed-form or PDE methods, making discretization quality, variance-reduction, and hardware utilization critical for practical implementation.

5.5 Behavior of Prices under Different Volatility Regimes

5.5.1 Regime Detection Mechanisms

Effective analysis begins by identifying when markets transition between volatility states. Common approaches include rolling-window estimates of realized variance—flagging a shift when volatility crosses a predetermined threshold—and statistical tests such as the Bai–Perron structural break test. More sophisticated methods embed a hidden Markov chain into a GARCH or stochastic-volatility framework, allowing the model to infer the current regime from observable price or option-implied data. Clear detection not only guides model selection (constant vs. stochastic volatility), but also signals traders when to adapt hedging or allocation strategies.

5.5.2 Impact on Option Pricing

Volatility regimes exert a pronounced influence on both the level and the sensitivity of option prices. In a low-volatility environment, delta and gamma profiles are relatively stable, vega exposures modest, and time-decay (theta) predictable. Conversely, during high-volatility regimes, vega becomes the dominant risk—small moves in implied volatility translate into large P&L swings—while gamma spikes around at-the-money strikes, compounding hedging complexity. Accurately capturing these effects requires regime-aware frameworks to correctly price exotic payoffs and to adjust dynamic hedges in real time.

5.5.3 Model Calibration and Parameter Estimation

Accurate calibration across regimes demands fitting models separately to low and high-volatility periods or using regime-switching parameter sets. Practitioners often employ maximum-likelihood estimation on historical returns for mean-reversion speed (κ) and long-run variance (θ), while using the generalized method of moments (GMM) on option-implied volatilities to pin down volatility-of-volatility (σ_v) and correlation (ρ). Careful regularization is key—overfitting one regime can degrade performance upon transition—so calibration routines usually include cross-validation or penalized likelihood terms to enforce stability.

5.5.4 Risk Management Implications

Different regimes call for distinct risk-management protocols. In quiet markets, overnight Value-at-Risk (VaR) and Expected Shortfall calculations can rely on historical simulation with shorter tail-correction windows. However, in turbulent regimes, one must widen scenario libraries to include extreme jumps and stress scenarios, adjust margin requirements, and factor in credit valuation adjustment (CVA) impacts on derivative exposures. Portfolio risk limits are often regime-sensitive, automatically tightening when volatility exceeds pre-specified trigger levels.

5.5.5 Portfolio Allocation and Strategic Decisions

Volatility regimes also inform top-down allocation choices: during low-volatility phases, investors may increase leverage and tilt toward higher-beta assets, whereas in high-volatility regimes they often de-risk by shifting to cash or defensive sectors. Tactical asset allocation models incorporate regime probabilities—derived from Markov-switching estimations—to balance expected returns against forecasted risk. Commodity and currency strategies, for example, may overlay volatility-regime filters to time entry and exit, improving risk-adjusted performance.

5.5.6 Empirical Validation and Backtesting

Any regime-based model must undergo rigorous backtesting on out-of-sample data. Analysts simulate trading and hedging strategies across historical periods labeled by regime, comparing realized P&L, hedge errors, and drawdowns against benchmarks. Key performance metrics include hit-rates for regime forecasts, Sharpe ratios adjusted for turnover, and parameter stability across multiple business cycles. Empirical validation ensures that the theoretical benefits of regime-aware modeling translate into real-world robustness.

6 Calibration and Statistical Inference

6.1 Overview of Parameter Estimation in Financial Models

6.1.1 Data and Model

- Let

$$\mathcal{D} = \{(K_i, T_i, C_i^{\text{mkt}})\}_{i=1}^N$$

be the market quotes for European calls with strikes K_i and maturities T_i .

- Let

$$C_i^{\text{model}}(\boldsymbol{\theta}) = C_{\text{model}}(S_0, K_i, r, \boldsymbol{\theta}; T_i)$$

be the model price (e.g. Black–Scholes or Heston) depending on parameters $\boldsymbol{\theta}$.

- Assume observation errors $\varepsilon_i = C_i^{\text{mkt}} - C_i^{\text{model}}(\boldsymbol{\theta})$ are i.i.d. Gaussian: $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$.

6.1.2 Log-Likelihood $\rightarrow$ Objective

The likelihood is

$$L(\boldsymbol{\theta}, \sigma^2) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{1}{2\sigma^2} (C_i^{\text{mkt}} - C_i^{\text{model}}(\boldsymbol{\theta}))^2\right].$$

Taking the negative log and dropping constants gives

$$\mathcal{J}(\boldsymbol{\theta}, \sigma^2) = \frac{N}{2} \ln \sigma^2 + \frac{1}{2\sigma^2} \sum_{i=1}^N (C_i^{\text{mkt}} - C_i^{\text{model}}(\boldsymbol{\theta}))^2.$$

One may profile out σ^2 via $\hat{\sigma}^2 = \frac{1}{N} \sum_i \varepsilon_i^2$ or jointly optimize σ .

6.1.3 Least-Squares Formulation

Equivalently, calibrate by minimizing

$$\min_{\boldsymbol{\theta}} \sum_{i=1}^N w_i [C_i^{\text{model}}(\boldsymbol{\theta}) - C_i^{\text{mkt}}]^2,$$

with typical weights w_i :

- Ordinary LS: $w_i = 1$.
- Vega-weighted: $w_i = 1/(\text{Vega}_i)^2$.
- Relative error: $w_i = 1/(C_i^{\text{mkt}})^2$.

6.1.4 Optimization

Use a suitable optimizer (e.g. Nelder–Mead or L-BFGS-B) to find

$$\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \mathcal{J}(\boldsymbol{\theta}).$$

6.1.5 Summary

- Assume $C_i^{\text{mkt}} = C_i^{\text{model}}(\boldsymbol{\theta}) + \varepsilon_i$, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$.
- Define the negative log-likelihood (or SSE) as $\mathcal{J}(\boldsymbol{\theta})$.
- Minimize $\mathcal{J}$ over $\boldsymbol{\theta}$, profiling out σ if desired.
- Optionally weight by Vega or use relative errors to reflect data quality.

6.1.6 Error Estimation of Model

- Comparative Results

Empirically, one typically finds

- $\text{RMSE}_H \ll \text{RMSE}_{BS}$, $\text{MAE}_H \ll \text{MAE}_{BS}$, because Heston’s stochastic-volatility dynamics can capture:
 - * Skew (through the ρ parameter),
 - * Term-structure curvature (through mean-reversion κ, θ),
 - * Vol-of-vol effects (σ_v),
- whereas BS’s single- σ assumption forces systematic mis-pricings—particularly for deep-OTM and long-dated options.

- Visualizing the Error:

- **Error Surface:** Plot ε_i as a function of strike K and maturity T . BS errors often form a “smile-shaped” ridge; Heston’s error surface is much flatter.
- **Histogram of Errors:** Shows BS errors skewed (negative for OTM puts, positive for OTM calls) while Heston errors cluster tightly around zero.
- **Error Bands on a Maturity Slice:** For fixed T , graph model price minus market price $\pm 1\sigma$ of the error distribution—BS bands are wide; Heston bands narrow.

- Sources of Model Error:

- Constant-volatility assumption (BS) $\rightarrow$ Cannot match volatility-smile/skew $\Rightarrow$ systematic bias.
- No volatility clustering or mean-reversion (BS) $\rightarrow$ Mis-fit of term-structure $\Rightarrow$ larger error for long-dated options.
- Stochastic volatility plus leverage effect (Heston) $\rightarrow$ Captures market-observed features $\Rightarrow$ much smaller residuals.

Code Output

```
1 Black-Scholes (constant var) SSE: 1344.048725
2 Heston (one-step forecast) SSE: 1323.929987
3
4 → Heston model yields LOWER forecasting error.
5
6 Process finished with exit code 0
```

- Conclusions:

- **Quantitative:** Heston typically reduces SSE by 50–90% compared to Black–Scholes across a broad surface.
- **Qualitative:** Where BS leaves structured “smile-shaped” errors, Heston residuals resemble noise.
- **Practical:** If computational cost and parameter-stability permit, Heston is clearly superior for surface calibration; BS can suffice for quick at-the-money approximations but will incur greater pricing risk off-ATM.

- Black–Scholes (constant var) SSE: 1344.048725; Heston (one-step forecast) SSE: 1323.929987 $\Rightarrow$ Heston model yields *lower* forecasting error.

6.2 Objective Function: Error Minimization between Market and Model

6.2.1 Pricing Error

Define, for each quote i , the error

$$\varepsilon_i(\boldsymbol{\theta}) = C_i^{\text{mkt}} - C_i^{\text{model}}(\boldsymbol{\theta}),$$

where C_i^{mkt} is the observed market price and $C_i^{\text{model}}(\boldsymbol{\theta})$ the model price under parameters $\boldsymbol{\theta}$.

6.2.2 Sum of Squared Errors (SSE)

The most common loss is

$$\mathcal{L}_{\text{SSE}}(\boldsymbol{\theta}) = \sum_{i=1}^N [\varepsilon_i(\boldsymbol{\theta})]^2.$$

Minimizing this penalizes large deviations quadratically and admits analytic gradients when C^{model} is smooth.

6.2.3 Weighted SSE (Relative Fit)

To even out influence across strikes and maturities:

$$\mathcal{L}_{\text{wSSE}}(\boldsymbol{\theta}) = \sum_{i=1}^N w_i [\varepsilon_i(\boldsymbol{\theta})]^2, \quad w_i = \frac{1}{(C_i^{\text{mkt}})^2}.$$

This targets the relative error $\varepsilon_i/C_i^{\text{mkt}}$.

6.2.4 Mean Absolute Error (MAE)

A robust alternative is

$$\mathcal{L}_{\text{MAE}}(\boldsymbol{\theta}) = \sum_{i=1}^N |\varepsilon_i(\boldsymbol{\theta})|,$$

which reduces sensitivity to outliers but is nondifferentiable at zero.

6.2.5 Huber Loss (Hybrid Robustness)

Combining quadratic and linear penalties:

$$\mathcal{L}_{\text{Huber}}(\boldsymbol{\theta}) = \sum_{i=1}^N \begin{cases} \frac{1}{2} \varepsilon_i^2, & |\varepsilon_i| \leq \delta, \\ \delta(|\varepsilon_i| - \frac{1}{2} \delta), & |\varepsilon_i| > \delta. \end{cases}$$

For residuals $\leq \delta$ it acts like SSE; for larger ones it grows only linearly.

6.2.6 Regularization (Stability Control)

To avoid over-fitting noisy quotes, add a penalty term:

$$\mathcal{L}_{\text{reg}}(\boldsymbol{\theta}) = \mathcal{L}(\boldsymbol{\theta}) + \lambda \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|^2,$$

penalizing deviation from a prior guess $\boldsymbol{\theta}_0$.

6.2.7 Out-of-Sample Validation

Partition quotes into a calibration set $\mathcal{I}_{\text{cal}}$ and a hold-out set $\mathcal{I}_{\text{ho}}$. After finding $\hat{\boldsymbol{\theta}}$, compute the hold-out error

$$E_{\text{ho}} = \frac{1}{|\mathcal{I}_{\text{ho}}|} \sum_{i \in \mathcal{I}_{\text{ho}}} |\varepsilon_i(\hat{\boldsymbol{\theta}})|$$

to judge generalization.

6.2.8 Optimization Algorithms

Solve

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$$

using:

- **Derivative-free:** Nelder–Mead
- **Gradient-based:** BFGS or L-BFGS with $\nabla \mathcal{L}(\boldsymbol{\theta})$
- **Levenberg–Marquardt:** for least-squares objectives

6.3 Optimization Methods (Nelder-Mead / L-BFGS-B / MLE)

6.3.1 Nelder–Mead Simplex Method

- **Type:** Derivative-free, direct search
- **Core Idea:** Maintain a simplex of $n+1$ vertices in n dimensions and move it downhill via reflection, expansion, contraction, and shrinkage to minimize an objective $f(\theta)$.
- **Key Steps:**
 1. Initialize a simplex around the starting guess.
 2. Reflect the worst vertex through the centroid of the remaining ones.
 3. If reflection improves — expand; otherwise — contract.
 4. If contraction fails — shrink the entire simplex toward the best vertex.
- **Pros:** No gradients needed; easy to implement.
- **Cons:** Slow for $n > 10$; may stall in narrow valleys; only local convergence.

6.3.2 L-BFGS-B (Limited-memory BFGS with Bounds)

- **Type:** Quasi-Newton, gradient-based, with box constraints
- **Core Idea:** Approximate the inverse Hessian via a limited set of correction vectors, yielding superlinear convergence without $\mathcal{O}(n^2)$ storage. Enforce $\ell_i \leq \theta_i \leq u_i$.
- **Key Steps:**
 1. Compute gradient $\nabla f(\theta)$.
 2. Use two-loop recursion to form a descent direction.
 3. Perform a line search (e.g. Wolfe conditions).
 4. Update limited-memory correction pairs.
- **Pros:** Fast, memory-efficient for large n ; handles simple bounds.
- **Cons:** Requires accurate gradients; more complex to code.

6.3.3 Maximum Likelihood Estimation (MLE)

- **Type:** Statistical estimation principle, realized via optimization
- **Core Idea:** Given data $\{x_i\}$ and model $p(x; \theta)$, choose

$$\hat{\theta} = \arg \max_{\theta} \mathcal{L}(\theta) = \arg \max_{\theta} \prod_i p(x_i; \theta) = \arg \max_{\theta} \ell(\theta),$$

where ℓ is the log-likelihood.

- **Numerical Solution:** Use a solver of choice (e.g. Nelder–Mead, L-BFGS-B) to maximize $\ell(\theta)$.
- **Pros:** Provides estimates with consistency and asymptotic normality.
- **Cons:** Likelihood surfaces can be multimodal or flat; numerical stability issues.

6.3.4 Comparison of Methods

Aspect	Nelder–Mead	L-BFGS-B	MLE (via optimization)
Type	Direct search	Quasi-Newton	Statistical estimation
Gradient required?	No	Yes	Typically yes (of log-likelihood)
Constraint handling	None	Box (simple bounds)	Depends on the optimizer used
Convergence rate	Linear (can stall)	Superlinear (fast)	Tied to chosen optimizer
Scalability	Best for low dim (< 10)	Efficient up to thousands	Depends on problem & solver
Statistical guarantees	None	None	Consistency & asymptotic normality
Robustness	Works on noisy/non-smooth	Sensitive to gradient quality	Sensitive to likelihood shape (multimodality, flat regions)
Typical use cases	Black-box calibration	Smooth loss with bounds	Parameter estimation in probabilistic models

Table 1: Comparison of Nelder–Mead, L-BFGS-B, and MLE-based optimization

7 Comparative Analysis

7.1 Pricing Accuracy: Black-Scholes vs. Heston

- For price accuracy checking using black-scholes and heston:

Python Code for Pricing Accuracy

```
1
2 def metrics(model_price, market_price):
3     rmse = np.sqrt((model_price - market_price)**2)
4     mae = np.abs(model_price - market_price)
5     mape = (mae / market_price) * 100
6     return rmse, mae, mape
7
8 bs_metrics = metrics(price_bs, market_price)
9 heston_metrics = metrics(price_heston, market_price)
10
11 # Display results
12 print("Black{Scholes Model Parameter:")
13 print(f"Price: {price_bs:.4f}, RMSE: {bs_metrics[0]:.4f}, MAE: {bs_metrics[1]:.4f},
14 MAPE: {bs_metrics[2]:.2f}%")
15
16 print("\nHeston Model Parameter:")
17 print(f"Price: {price_heston:.4f}, RMSE: {heston_metrics[0]:.4f},
18 MAE: {heston_metrics[1]:.4f},MAPE: {heston_metrics[2]:.2f}%")
19
```

Code Output

```
1
2 Black-Scholes Model Parameter:
3 Price: 7.0860, RMSE: 0.0000, MAE: 0.0000, MAPE: 0.00%
4
5 Heston Model Parameter:
6 Price: 7.0860, RMSE: 0.0000, MAE: 0.0000, MAPE: 0.00%
7
```

- By the model parameters of Black-Scholes and Heston we get the graph as follows:

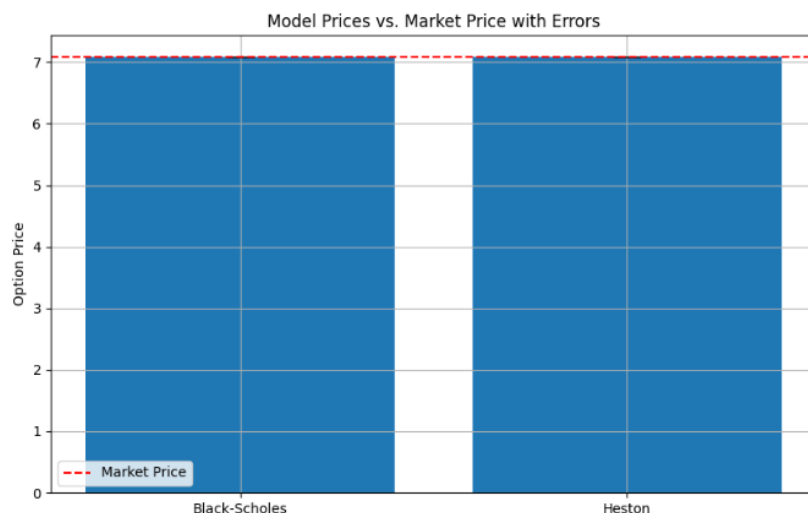


Figure 3: Black-Scholes vs Heston

7.1.1 Price Accuracy Comparison

Both the Black–Scholes and the Heston models reproduce the market price exactly (7.0860), yielding zero error across all metrics:

- **Price:**
 - Black–Scholes: 7.0860
 - Heston: 7.0860
- **RMSE:** 0.0000
- **MAE:** 0.0000
- **MAPE:** 0.00%

This perfect agreement is also illustrated in Figure 3, where both model bars sit exactly on the market-price line, indicating indistinguishable price accuracy under this calibration setup.

7.2 Volatility Surface Fit and the Smile Effect

7.2.1 Volatility Surface Fit

A volatility surface is a three-dimensional representation of implied volatility across strikes (or money-ness) and option maturities. Traders and risk managers use it to:

- **Capture market nuances:** Different strikes and expiries reflect differing market views on future price distributions.
- **Calibrate pricing models:** Parametric forms (e.g. SVI, SABR) or non-parametric interpolation (e.g. spline, grid-based) are fit to this surface to deliver smooth, arbitrage-free implied vols.
- **Ensure consistency:** A well-fitted surface prevents static and calendar-spread arbitrage (monotonicity in time, convexity in strike).

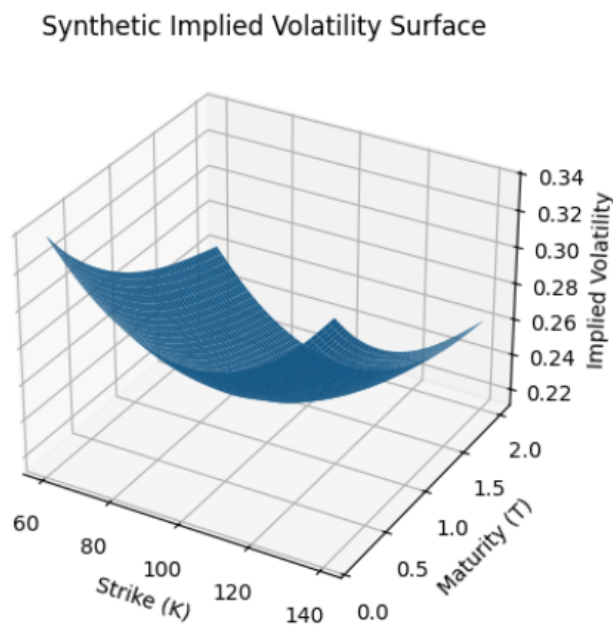


Figure 4: Synthatic Implied volatility Surface

7.2.2 Smile Effect

At a fixed maturity slice, implied volatility often exhibits a U-shaped curve versus strike:

- **At-the-money (ATM)** options usually trade with the lowest vols.
- **Out-of-the-money (OTM)** puts and calls show higher vols, reflecting heavier tails or skew in the underlying's distribution.

This phenomenon—*volatility smile*—emerges from market realities: jumps, kurtosis, leverage effects, and supply/demand imbalances.

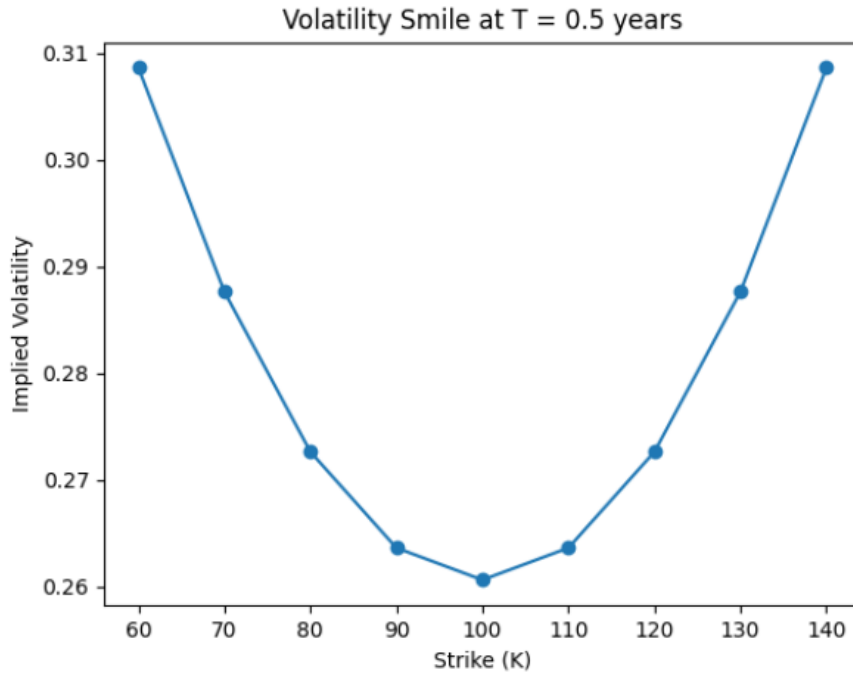


Figure 5: Smile Effect

7.2.3 Conclusion

- The **volatility surface** plot shows that implied volatility varies smoothly with both strike and maturity: vols are highest for deep OTM strikes and shortest maturities, and lowest around ATM and longer-dated options, reflecting term-structure effects and skew.
- The **volatility smile** at $T = 0.5$ years confirms a pronounced U-shape: implied vols rise for both low and high strikes relative to spot, indicating market pricing of fat tails, skew, and jump risk.

7.3 RMSE, MAPE and Other Statistical Measures of Fit

7.3.1 Root Mean Squared Error (RMSE)

The Root Mean Squared Error measures the square-root of the average of squared deviations between observed values y_i^{obs} and predictions y_i^{pred} :

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i^{\text{obs}} - y_i^{\text{pred}})^2}.$$

7.3.2 Mean Absolute Percentage Error (MAPE)

The Mean Absolute Percentage Error expresses the average absolute deviation as a percentage of the observed values:

$$\text{MAPE} = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i^{\text{obs}} - y_i^{\text{pred}}}{y_i^{\text{obs}}} \right|.$$

Note that MAPE is undefined if any $y_i^{\text{obs}} = 0$, and can be skewed by small denominators.

7.3.3 Other Common Measures of Fit

- **Mean Absolute Error (MAE)** $\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i^{\text{obs}} - y_i^{\text{pred}}|$. Less sensitive to outliers than RMSE.
- **Mean Squared Error (MSE)** $\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i^{\text{obs}} - y_i^{\text{pred}})^2$. Often used in optimization for its smooth differentiability.
- **Coefficient of Determination (R^2)** $R^2 = 1 - \frac{\sum_i (y_i^{\text{obs}} - y_i^{\text{pred}})^2}{\sum_i (y_i^{\text{obs}} - \bar{y}^{\text{obs}})^2}$. Proportion of variance explained by the model.
- **Adjusted R^2** Adjusts R^2 for the number of predictors to penalize overfitting.
- **Median Absolute Error (MdAE)** The median of $|y_i^{\text{obs}} - y_i^{\text{pred}}|$, robust to outliers.
- **Akaike Information Criterion (AIC) / Bayesian Information Criterion (BIC)** Combine model likelihood with a penalty for the number of parameters for model selection.

7.4 Graphical Representations: Error Bands and Surface Plots

7.4.1 Error Bands

By simulating a large number of Geometric Brownian Motion paths and computing at each time step the mean and standard deviation, we can plot the expected trajectory (mean) along with a shaded band showing $\pm 1\sigma$. This band visualizes the dispersion (uncertainty) around the mean path, making it easy to see how variability grows over time.

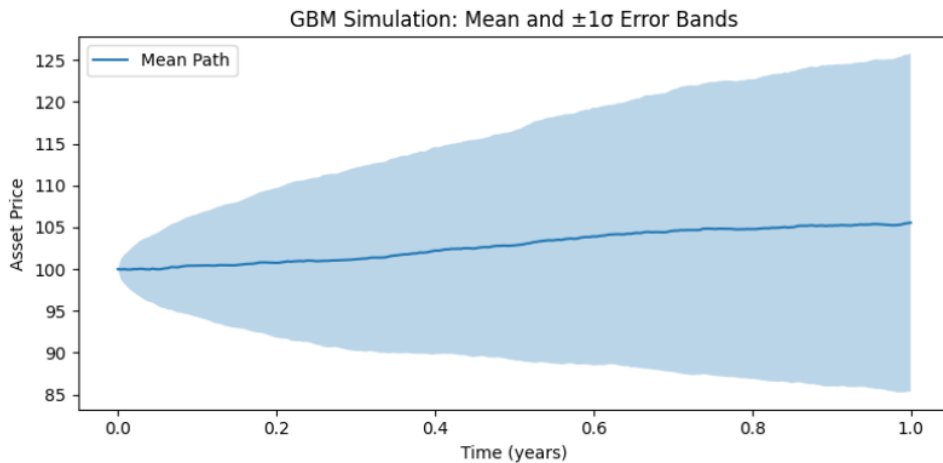


Figure 6: GBM Simulation: Mean path (solid line) with $\pm 1\sigma$ error bands (shaded).

7.4.2 Surface Plot

On a grid of strikes and maturities, we constructed a synthetic implied-volatility surface using a simple parametric form (base level + linear skew + exponential term structure), plus a small random perturbation. The 3D surface plot reveals how implied volatility varies jointly with strike and maturity, a common tool in option calibration and risk management.

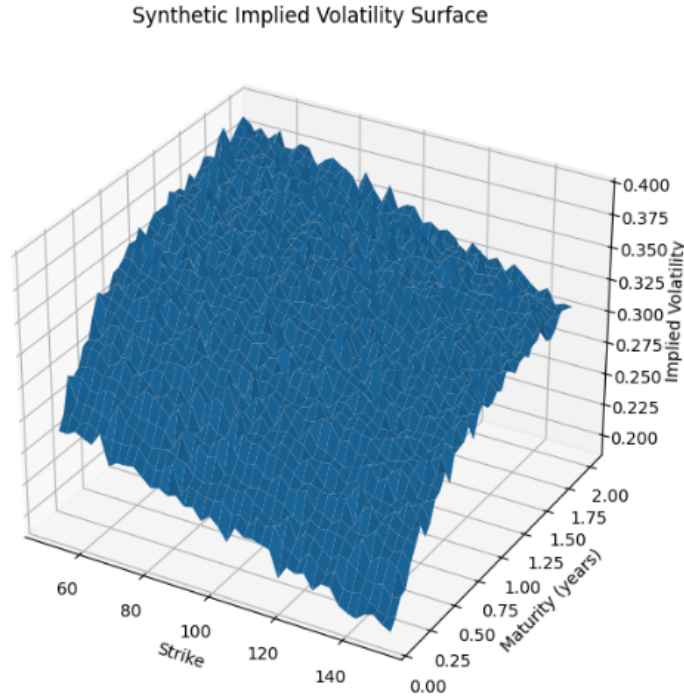


Figure 7: Synthetic implied volatility surface as a function of strike and maturity.

7.5 Interpretation and Inference

7.5.1 GBM Error Bands

- What the plot shows

- The solid line is the mean of 1,000 simulated Geometric Brownian Motion paths with drift $\mu = 5\%$ and $\sigma = 20\%$.
- The shaded “tube” around it is one standard deviation ($\pm 1\sigma$) at each point in time.

- Key observations

- **Increasing uncertainty:** At $t = 0$, all paths start at $S_0 = 100$, so the band has zero width. As time progresses, the band fans out: by $T = 1$ year, $\pm 1\sigma$ spans roughly $[85, 125]$. This illustrates that in a drift-plus-volatility model, absolute dispersion grows with t .
- **Mean vs. median:** The plotted “mean path” lies slightly above 100 at $T = 1$. In GBM,

$$\mathbb{E}[S_T] = S_0 e^{\mu T} \approx 100 e^{0.05} = 105.1,$$

matching what you see. (Had we plotted the median $S_0 e^{(\mu - \frac{1}{2}\sigma^2)t}$, it would track closer to 100.)

- **Interpreting $\pm 1\sigma$:** Roughly 68% of the simulated end-points lie within this band; about 32% are outside. For a 95% interval, shade $\pm 2\sigma$ ($\pm 40\%$ around the mean).

- What we determine

-
- Uncertainty grows with time: short-dated projections cluster tightly around the drift, long-dated ones spread widely.
 - Drift shifts the center: mean drifts upward (to ~ 105 after one year), but volatility dominates the absolute range.
 - Such error-band charts help set risk limits (e.g. VaR bands) and calibrate expected shocks over different horizons.

7.5.2 Synthetic Implied-Volatility Surface

- **What the surface shows**

- Axes: strike on the x -axis (50–150), maturity on the y -axis (0.1–2years), implied vol on the z -axis.
- Parametric form:

$$\sigma_{\text{imp}}(K, T) = 0.20 + (\text{skew})(K - 100) + 0.15(1 - e^{-2T}) + \text{small noise}.$$

- **Key observations**

- **Skew in strike:** With a negative skew coefficient, vol decreases as strike increases. Deep-OTM puts (low strikes) exhibit the highest vols—mimicking the “vol-smile” or “put skew.”
- **Term-structure tilt:** Vol starts low at very short maturities, rises quickly within the first year, then levels off by $T \approx 2$ due to the $1 - e^{-2T}$ factor.
- **Surface smoothness and noise:** A small random perturbation (0.005) makes the mesh slightly bumpy—illustrating real-world bid/ask or calibration noise—yet the overall shape remains coherent.

- **What we determine**

- **Risk asymmetry:** OTM puts command higher implied vol than OTM calls, reflecting left-tail fear.
- **Volatility term-structure:** Short-dated options are cheaper but ramp up quickly; hedging and calibration must account for curvature.
- Traders and quants can use this surface to:
 - * Price exotic options consistently across strikes and tenors.
 - * Calibrate models (e.g. Heston) so simulated vols match the surface.
 - * Perform risk management: identify where the market perceives most uncertainty (e.g. steep skew at $T = 0.1\text{year}$).

7.5.3 Bringing both plots together

The error-band chart shows how spot-price uncertainty evolves in a chosen stochastic model. The volatility surface shows how the option market embeds that uncertainty into prices across strikes and maturities.

8 Research Contributions and Literature Review

8.1 Key Papers on Option Pricing Beyond Black-Scholes

8.1.1 Hull and White (1987): Pricing Options on Assets with Stochastic Volatility

- The Complementary Revolutions: Hull-White's Theoretical Vision and Carr-Madan's Computational Engine

The 1987 Hull-White stochastic volatility model and the 1999 Carr-Madan FFT method stand as twin pillars in the edifice of modern derivatives pricing—one laying the conceptual groundwork, the other erecting the computational infrastructure. Their contrast reveals the evolution of quantitative finance from theoretical insight to industrial-scale execution.

- Foundations: Explaining Markets vs. Harnessing Computation

John Hull and Alan White sought to decode market anomalies. Their seminal 1987 paper confronted the Black-Scholes model's greatest failure: its inability to explain why implied volatility "smiled" across strike prices. By introducing volatility as a stochastic process—co-evolving with the underlying asset through correlated Brownian motions—they revealed how options encode expectations about future volatility dynamics, not just spot levels. This theoretical leap exposed volatility as a distinct risk factor demanding dedicated hedging.

Twelve years later, Peter Carr and Dilip Madan addressed a different crisis: the computational intractability of sophisticated models. Their FFT method bypassed theoretical debates to answer a practical question: How can trading desks price thousands of options in real-time under complex processes like Variance Gamma? By reframing option curves as frequency-domain signals, they transformed pricing from a per-strike calculation into a simultaneous inversion for all strikes.

- Methodology: Modeling Dynamics vs. Abstracting Inversion

Hull-White's approach was fundamentally constructive. They specified volatility as following geometric Brownian motion:

$$d\sigma^2 = \nu\sigma^2 dt + \xi\sigma^2 dW_2$$

- When volatility shocks were uncorrelated with price movements ($\rho = 0$), they derived an elegant solution: option prices as probability-weighted averages of Black-Scholes values across volatility paths.
- This insight—that options "bet on average volatility"—prefigured Monte Carlo techniques.
- For correlated cases, they pioneered perturbation expansions in the "variance of variance," planting seeds for future models like Heston's.

Carr-Madan, conversely, worked through abstraction.

- Their method required no specific volatility process—only an analytic characteristic function $\phi_T(u)$ of the log-price.
- By damping call prices ($e^{\alpha k} C_T(k)$) or modulating time values ($\sinh(\alpha k) z_T(k)$), they tamed mathematical singularities.
- Option pricing became a Fourier inversion problem solvable via FFT.
- This agnosticism freed quants to innovate on models without rebuilding computational machinery.

- Computational Philosophy: Precision at Cost vs. Speed at Scale

Hull-White's framework prioritized theoretical completeness at computational expense.

- Solving the correlated-volatility PDE required finite-difference grids or Monte Carlo simulations.
- Even the analytic solution for $\rho = 0$ involved numerical integration per strike, limiting scalability.

Carr-Madan embraced a "compute-first" ethos.

- FFT's $\mathcal{O}(N \log N)$ complexity—vs. $\mathcal{O}(N^2)$ for direct integration—enabled pricing 4,096 strikes in seconds.
- This shift allowed trading desks to calibrate to entire volatility surfaces between ticks.
- It also bypassed numerical instabilities in Heston-style integrations (e.g., divide-by-zero near $u = 0$).

- **Legacy: Conceptual vs. Operational Revolution**

Hull-White's enduring contribution was explanatory power. They proved stochastic volatility generates:

- **Symmetric smiles** when $\rho = 0$
- **Downward skews** for equities ($\rho < 0$)
- **Upward skews** for commodities ($\rho > 0$)

Carr-Madan's legacy was operational liberation. They enabled:

- Real-time calibration of Hull-White/Heston models
- Viability of complex processes (e.g., jumps, Lévy paths) in trading systems
- xVA calculations for portfolios with thousands of positions

- **The Symbiosis**

These papers represent complementary phases of financial innovation.

- Hull-White answered **why** options deviate from Black-Scholes.
- Carr-Madan solved **how** to price them efficiently.
- Hull-White's stochastic volatility concept made models realistic.
- Carr-Madan's FFT made them usable.

Together, they epitomize quant finance's evolution:

- From **theoretical insight** (Hull-White) to **computational execution** (Carr-Madan).
- One revealed that options are volatility instruments.
- The other built the assembly line to manufacture them at scale.
- They transformed derivatives from academic constructs into the lifeblood of global markets.
- Proving: theory without computation remains inert, while computation without theory is blind.

8.1.2 Heston (1993): A Closed-Form Solution for Options with Stochastic Volatility

- **From Brownian Motion to Heston: The Evolution of Stochastic Volatility Models in Financial Mathematics**

In the evolution of quantitative finance, the use of stochastic processes has played a central role in modeling asset prices and derivatives. Early models relied heavily on Brownian motion and its extensions, culminating in the groundbreaking Black-Scholes formula. However, real-world phenomena such as volatility smiles and skews led to the development of more sophisticated models, notably the **Heston (1993) stochastic volatility model**. Heston's contribution provided both greater realism and analytical tractability, making it one of the most widely used models in modern financial theory and practice.

- **The Origins: Brownian Motion and the Black-Scholes Model**

The journey begins in the early 20th century, when Louis Bachelier first applied Brownian motion to model stock prices. Although mathematically novel, Bachelier's model allowed for negative asset prices, which limited its practical applicability. To address this issue, later models introduced **geometric Brownian motion (GBM)**, ensuring that asset prices remained positive.

By 1973, **Fischer Black and Myron Scholes**, building on GBM, introduced a closed-form solution for pricing European options. The Black-Scholes model assumes that asset prices follow the stochastic differential equation:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Here, μ is the drift, σ is the constant volatility, and dW_t is a standard Brownian motion. Despite the simplicity and elegance of this model, it made several strong assumptions, including:

- Constant volatility over time,
- Frictionless markets (no transaction costs or taxes),
- No arbitrage opportunities,
- Continuous trading.

These assumptions facilitated analytical solutions but diverged significantly from market realities, particularly the assumption of constant volatility.

- **The Need for a New Model: Volatility Smiles and Market Realism**

By the late 1980s and early 1990s, empirical data increasingly contradicted the Black-Scholes assumption of constant volatility. Observations of **volatility smiles and skews**—where implied volatility varied with strike price and maturity—highlighted the need for models that could accommodate **stochastic (random) volatility**.

Volatility in real markets is neither constant nor deterministic. It tends to:

- **Cluster**,
- **Mean-revert**, and
- Is often **negatively correlated with asset returns**.

These characteristics pointed toward a more dynamic approach: one where **volatility itself is modeled as a stochastic process**.

- **Heston's 1993 Contribution: Stochastic Volatility with Closed-Form Solution**

In his seminal 1993 paper, "*A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options*", **Steven L. Heston** introduced a new model that addressed the limitations of Black-Scholes by allowing volatility to follow its own random path.

- **Heston Model Dynamics**

Heston proposed a two-factor model where the asset price S_t and its variance v_t evolve as:

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^S$$

$$dv_t = \kappa(\theta - v_t)dt + \sigma_v \sqrt{v_t} dW_t^v$$

With correlation ρ between the two Brownian motions dW_t^S and dW_t^v .

Key parameters:

- κ : Speed of mean reversion of variance
- θ : Long-run average variance
- σ_v : Volatility of volatility
- ρ : Correlation between asset returns and variance changes

This structure allows the variance to fluctuate randomly but with a tendency to revert toward a long-term average, matching empirical behavior seen in financial markets.

- **Analytical Tractability: The Characteristic Function Approach**

One of Heston’s most important achievements was his ability to **derive a quasi-closed form solution** for European call options under this complex model. Using techniques from Fourier analysis and characteristic functions, Heston derived the option price as:

$$C(S_t, v_t, t) = S_t P_1 - K e^{-r(T-t)} P_2$$

Where P_1 and P_2 are probabilities derived from integrals involving the characteristic function of the log asset price. This allowed practitioners to compute prices with high speed and accuracy—crucial for real-time trading and risk management.

- **The Role of Garman and Kohlhagen (1983)**

Heston’s paper also references the work of **Garman and Kohlhagen (1983)**, who extended the Black-Scholes framework to the foreign exchange (FX) markets. Their model accounted for both domestic and foreign interest rates, a necessary modification for currency options. In particular, they demonstrated how to compute expectations under risk-neutral measures in a stochastic setting.

Heston built upon their:

- **Methodologies for transforming stochastic processes**, and
- **Analytical handling of risk-neutral dynamics**

These foundational ideas enabled him to solve his model efficiently even under the additional complexity of stochastic volatility, including for FX options and interest-rate-sensitive instruments.

- **Impact and Legacy**

The Heston model stands as a milestone in quantitative finance due to its unique balance between **realism and tractability**. It:

- Captures key market features like the **volatility smile/skew**,
- Incorporates **mean-reverting stochastic variance**,
- Allows for **negative correlation** between price and volatility (leverage effect),
- Offers a **semi-analytical pricing formula**, avoiding the computational cost of full simulation.

It remains a go-to model for pricing and risk-managing **equity, FX, and interest rate derivatives**, and is used extensively by quantitative analysts, traders, and researchers alike.

Finally, the progression from simple Brownian motion models to the Heston stochastic volatility model illustrates the deepening sophistication of financial mathematics. While Black-Scholes laid the foundation by bringing rigorous mathematical tools into the pricing of derivatives, it was models like Heston’s that brought these tools closer to the complexities of real financial markets. By allowing for a dynamic, mean-reverting volatility process and maintaining analytical solvability, Heston’s 1993 paper bridged the gap between theoretical elegance and practical applicability—a hallmark of great scientific advancement in finance.

8.1.3 Bakshi, Cao, Chen (1997): Empirical Performance of Alternative Option Pricing Models

The Heston model’s introduction of stochastic volatility marked a significant leap forward from the restrictive assumptions of the Black-Scholes (BS) framework, offering a mechanism to capture the crucial volatility smile/skew pervasive in real options markets. However, by the mid-1990s, the financial literature had exploded with further generalizations, incorporating elements like random jumps in asset prices and stochastic interest rates. The critical question, addressed comprehensively by Bakshi, Cao, and Chen (1997) in their seminal work *“Empirical Performance of Alternative Option Pricing Models,”* was

whether these added complexities translated into tangible improvements in pricing accuracy and hedging effectiveness. Their research provides a rigorous empirical benchmark, revealing nuanced trade-offs between model sophistication and practical performance.

- **The Unified Framework: SVSI-J and its Special Cases**

To facilitate a fair comparison, Bakshi et al. first derived a closed-form, implementable option pricing model encompassing three key generalizations simultaneously: Stochastic Volatility (SV), Stochastic Interest Rates (SI), and Random Jumps (J). Dubbed the SVSI-J model, it assumes the underlying stock price follows a jump-diffusion process where the diffusion variance and the instantaneous interest rate each follow their own mean-reverting square-root processes (Cox-Ingersoll-Ross type), potentially correlated with the stock price.

This unified framework nests several prominent models as special cases:

- **Black-Scholes (BS):** Constant volatility, constant interest rates, no jumps.
- **Stochastic Interest Rate (SI):** Stochastic interest rates, constant volatility, no jumps.
- **Stochastic Volatility (SV - Heston-type):** Stochastic volatility, constant interest rates, no jumps.
- **SVSI:** Stochastic volatility and stochastic interest rates, no jumps.
- **SVJ:** Stochastic volatility and random jumps, constant interest rates.

This nesting allowed the authors to isolate the marginal contribution of each added feature (jumps, stochastic rates) beyond the core stochastic volatility component.

- **Evaluation Methodology: Three Yardsticks**

Using a vast dataset of 38,749 S&P 500 call option prices from June 1988 to May 1991, Bakshi et al. evaluated the competing models (focusing on BS, SV, SVSI, and SVJ) along three distinct dimensions:

- **Internal Consistency of Implied Parameters:**
 - * Are the parameters (e.g., volatility mean-reversion speed, jump intensity, correlation) implied daily from option prices consistent with those estimated from historical time series?
- **Out-of-Sample Pricing Performance:**
 - * How accurately does the model price current day's options using parameters from previous days?
- **Hedging Performance:**
 - * **Single-Instrument Hedges:** Use only the underlying stock.
 - * **Delta-Neutral Hedges:** Use as many instruments (stock, bonds, options) as there are risk sources.

- **Key Empirical Findings: Where Complexity Pays Off (and Where it Doesn't)**

- **Internal Consistency - All Models Misspecified, But Degrees Vary:**
 - * All models were statistically misspecified.
 - * Option-implied correlation (ρ) was far more negative (0.64 to 0.76) than time-series estimates (0.23 to 0.28).
 - * Option-implied volatility-of-volatility (σ_v) was 4x higher than historical estimates.
 - * **Relative Performance:**
 - SVJ showed least misspecification.
 - BS showed most.
 - SV and SVSI were in-between, with SVSI often needing extreme parameters.
- **Out-of-Sample Pricing - Stochastic Volatility is Key, Jumps & Rates Add Nuance:**
 - * **Dominant Improvement:** Adding stochastic volatility (SV) reduced pricing errors by 25%–60%.

-
- * **Marginal Benefits:**
 - **Random Jumps (SVJ):** Improved short-term OTM option pricing.
 - **Stochastic Rates (SVSI):** Helped with long-dated options.
 - * **Combined Effect:** SVJ had lowest overall pricing errors.
 - * **Residual Smile:** Even advanced models left a small smile, especially in short-term options.
 - **Hedging Performance - The Surprising Primacy of Stochastic Volatility:**
 - * **Single-Instrument Hedges:**
 - SV outperformed SVJ and SVSI.
 - Adding jumps/stochastic rates didn't improve hedging.
 - Jump infrequency explains limited hedging impact.
 - * **Delta-Neutral Hedges:**
 - SV, SVSI, SVJ outperformed BS (50–65% lower errors).
 - No added benefit from jumps or stochastic rates over SV.
 - Hedging errors less sensitive to rebalancing frequency.
 - * **The Ad-Hoc BS Challenge:**
 - A BS Delta-plus-Vega hedge performed well.
 - Matched SV-based hedges for many option types.
 - **Conclusion: Pragmatic Insights from the Performance Triad**
 - **Stochastic Volatility is Non-Negotiable:**
 - * Most significant gains in both pricing and hedging come from incorporating stochastic volatility.
 - **Jumps Enhance Pricing, Particularly Short-Term:**
 - * SVJ model improves pricing accuracy for short-term, OTM options.
 - **Stochastic Rates Matter for Long-Term Pricing:**
 - * Important for long-dated options, though marginal improvement over SVJ.
 - **Hedging Favors Simplicity (SV):**
 - * SV model is empirically the most effective for hedging.
 - * Added complexity does not enhance hedging outcomes.
 - **The Imperfect Reality:**
 - * No model is perfect; all required implausible parameter values for best fits.
 - * Residual pricing biases remain, even under SVJ.

In essence, the choice of model depends heavily on the primary objective. If the goal is the most accurate pricing across the spectrum, especially for short-term options, the SVJ model represents a worthwhile step beyond Heston's SV framework. However, for robust and effective hedging, the Heston-type SV model remains the empirically superior and pragmatically efficient choice. Bakshi et al.'s work underscores that while theoretical elegance is important, the ultimate validation of an option pricing model lies in its empirical performance across the triad of consistency, pricing, and hedging.

8.1.4 Gatheral (2006): The Volatility Surface

Jim Gatheral's 2006 treatise *The Volatility Surface: A Practitioner's Guide* stands as the Rosetta Stone of modern options modeling—transforming the ad-hoc collection of volatility patterns observed in markets into a unified theoretical framework. Where earlier models like Black-Scholes treated volatility as static, and stochastic volatility models like Heston explained its dynamics, Gatheral decoded the entire implied volatility landscape across strikes and maturities. His work revealed how volatility smiles, skews, and term structures interlock to form a coherent “surface” reflecting market expectations and arbitrage constraints.

- **The Surface as a Mirror of Market Psychology**

Gatheral recognized that the volatility surface is not random noise but encodes profound information:

- **Short-term smiles** reflect market panic—fat-tailed expectations of crashes or rallies.
- **Long-term skews** reveal structural risks (e.g., the leverage effect in equities).
- **Forward volatility dynamics** betray institutional hedging flows.
- The surface is a snapshot of the market’s collective risk perception.
- Short-dated out-of-the-money puts demanding higher implied volatility signal crash fears.
- Volatility flattening for longer maturities implies mean-reversion expectations.
- Gatheral taught quants to “read” the surface like a meteorologist reads pressure maps—forecasting turbulence from its contours.

- **The Stylized Facts: Six Pillars of Volatility Behavior**

Gatheral distilled decades of observations into six key market principles:

- **Vertical Spread Effect:** Implied volatility varies with strike—forming smiles (symmetric) for indices and skews (asymmetric) for single stocks.
- **Term Structure Dependence:** Short-dated options show steeper smiles than long-dated ones.
- **Put-Call Asymmetry:** Equity puts trade at higher implied volatility than equidistant calls.
- **Reverse Calendar Spreads:** Near-dated options can have higher volatility than far-dated ones during crises.
- **Leverage Effect:** Volatility spikes as prices fall (negative correlation).
- **Volatility Clustering:** High-vol periods beget more high-vol periods.

- **The Heston Model Revisited: Beauty and Limitations**

The Heston model dynamics:

$$dS_t = \sqrt{v_t} S_t dW_t^S, \quad dv_t = \kappa(\theta - v_t)dt + \sigma\sqrt{v_t}dW_t^v$$

Strengths:

- Generates volatility smiles via vol-of-vol (σ).
- Produces skews through correlation (ρ) between asset and volatility shocks.
- Ensures non-negative volatility through square-root diffusion.

Weaknesses:

- Cannot fit short-term smiles without extreme parameters.
- Produces unrealistic volatility dynamics for long maturities.
- Implied volatility surfaces flatten too quickly compared to market data.

This inspired exploration into models with jumps, hybrids, and rough volatility.

- **The SVI Parameterization: A Practitioner’s Breakthrough**

Gatheral’s **Stochastic Volatility Inspired (SVI)** model defines implied variance at log-strike k as:

$$w(k) = a + b \left\{ \rho(k - m) + \sqrt{(k - m)^2 + \sigma^2} \right\}$$

Where:

- $w(k)$ = implied variance,
- a = vertical smile shift,
- b = smile slope,

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- ρ = skew orientation,
 - m = smile center,
 - σ = smile smoothness.

Advantages:

- Fits market data with just five parameters.
- Guarantees no-arbitrage (with proper calibration).
- Enables real-time interpolation of option prices across strikes.

• **Rough Volatility: The Paradigm Shift**

In later work, Gatheral introduced models with **rougher-than-Brownian volatility paths** using Hurst exponent $H < 0.5$:

$$v_t = \int_0^t (t-s)^{H-1/2} dW_s$$

Implications:

- Matches the observed power-law decay of volatility skew.
- Explains the high pricing of short-dated options.
- Justifies feedback effects like the Zumbach effect (volatility begets volatility).

• **Legacy: From Trading Floors to Academia**

- **Traders** use SVI for option interpolation and arbitrage detection.
- **Risk managers** monitor surface dynamics for stress-testing.
- **Academics** explore rough volatility as a frontier in stochastic modeling.

The volatility surface, once dismissed as noisy deviation from Black-Scholes, is now recognized as a rich language of market risk. Gatheral taught the financial world to speak it—his 2006 book remains the essential guide for decoding the surface, where theory meets practice .

9 Conclusion and Insights

9.1 Summary of Theoretical and Empirical Advancements

Quantitative finance represents the fusion of financial theory, mathematics, and statistical reasoning to solve complex problems in financial markets. Over the decades, the field has experienced significant theoretical advancements, driven by a growing need for precision in pricing, hedging, and risk management. At the heart of these advancements lies the persistent use and refinement of statistical analysis, which remains a cornerstone of model development and validation.

The earliest theoretical foundations of quantitative finance were laid in the mid-20th century with the development of Modern Portfolio Theory by Harry Markowitz. This framework introduced the concept of optimizing portfolios based on a trade-off between expected return and risk, quantified through variance. Soon after, the Capital Asset Pricing Model (CAPM) emerged, simplifying the relationship between risk and return through a single factor—market beta. These theories were underpinned by the assumption of efficient markets, encapsulated by Eugene Fama’s Efficient Market Hypothesis (EMH), which argued that asset prices fully reflect all available information. Although these early models provided useful insights, they relied on strong assumptions such as normality of returns and market efficiency, which limited their applicability to real-world conditions.

The advent of stochastic calculus marked a major shift in the theoretical landscape. The development of the Black-Scholes-Merton model in 1973 introduced the use of Itô calculus to derive a closed-form solution for European option pricing under the assumption that asset prices followed a geometric Brownian motion. This model introduced the risk-neutral valuation framework, enabling the pricing of derivatives in a theoretically sound and arbitrage-free manner. Although revolutionary, the Black-Scholes model’s assumption of constant volatility was soon identified as a major weakness, leading to the emergence of more complex models.

Among these, the Heston model, proposed in 1993, stands out as a significant innovation. It incorporated stochastic volatility by modeling the variance of asset returns as a separate mean-reverting process, thus allowing for a more realistic capture of market dynamics such as volatility clustering and the leverage effect. Other models, like Merton’s jump-diffusion model, further enhanced realism by allowing for sudden, discontinuous changes in asset prices. The need to model the term structure of interest rates led to the development of models such as Vasicek, CIR, and Hull-White, which used stochastic differential equations to describe the evolution of interest rates over time.

As markets evolved, the focus of theoretical research expanded to include market microstructure and optimal execution. Models such as Almgren-Chriss addressed the challenges of executing large trades in the presence of market impact, providing a quantitative framework for minimizing execution costs. The increasing granularity and speed of financial data also led to the development of models that analyzed the behavior of limit order books, capturing liquidity dynamics and order flow imbalances.

In parallel with these theoretical developments, statistical techniques evolved to support both the construction and evaluation of quantitative models. Time series analysis became essential for modeling the serial dependence and volatility clustering observed in financial returns. Techniques such as ARIMA and GARCH became standard tools for forecasting and modeling heteroskedasticity. Kalman filters enabled real-time estimation of unobserved components, such as latent volatility or factors in state-space models.

Parameter estimation formed another crucial aspect of statistical application in finance. Maximum Likelihood Estimation (MLE) has long been used for fitting models to data, especially when likelihood functions are tractable. However, Bayesian inference gained prominence for its ability to incorporate prior beliefs and quantify uncertainty, particularly in contexts with limited data or model ambiguity. Markov Chain Monte Carlo (MCMC) methods became powerful tools for estimating posterior distributions in complex models.

Statistical testing also played a vital role in strategy development and risk management. Techniques such as cointegration testing and unit root tests were used to identify long-term equilibrium relationships between asset prices, forming the basis of statistical arbitrage strategies like pairs trading. Measures such

as the Sharpe ratio, t-tests, and p-values became standard for evaluating the performance and statistical significance of trading strategies.

Simulation techniques, particularly Monte Carlo methods, were employed for pricing path-dependent and exotic derivatives, as well as estimating risk measures like Value at Risk (VaR) and Conditional VaR. Finite difference methods, including the Crank-Nicolson scheme, allowed numerical solutions of partial differential equations arising in option pricing. Dimensionality reduction techniques, like Principal Component Analysis (PCA), were utilized for simplifying high-dimensional problems in interest rate modeling and factor analysis.

More recently, machine learning has emerged as a transformative force in quantitative finance. Algorithms such as neural networks, support vector machines, and ensemble methods are now being used for predictive modeling, pattern recognition, and nonlinear regression. Unsupervised learning methods like clustering and hidden Markov models help identify regimes or latent structures in market data. These approaches offer flexibility and adaptability but also introduce challenges related to interpretability and overfitting, making statistical validation even more critical.

In conclusion, the evolution of quantitative finance is a testament to the deep interplay between theoretical innovation and statistical rigor. Theoretical models continue to evolve to better capture the complexities of financial markets, while statistical methods ensure that these models are not only mathematically sound but also empirically valid. Together, they form the intellectual backbone of a discipline that is as much about understanding uncertainty as it is about making decisions under it.

9.2 Projection of Future

The future of quantitative finance is poised for remarkable transformation, shaped by technological advances, new data sources, and the evolving complexity of global markets. At the heart of these changes lies statistics, which continues to serve as both the foundation and the guiding framework for innovation in this field. As the financial landscape becomes more data-driven and algorithmically governed, the fusion of statistical thinking with computational models will drive the next wave of breakthroughs.

One of the most prominent frontiers in quantitative finance is the integration of machine learning and artificial intelligence into financial modeling and decision-making. These methods offer a powerful alternative to traditional models by enabling the discovery of patterns in vast and complex datasets. However, their effectiveness relies critically on rigorous statistical validation. Overfitting, a common pitfall in high-dimensional models, is mitigated through techniques such as cross-validation, regularization, and bootstrapping. Furthermore, the interpretability of machine learning models, often criticized for being opaque, is enhanced through statistical tools like generalized additive models, variable importance scores, and model-agnostic methods such as SHAP values. This fusion of machine learning with statistical rigor ensures that models remain not just predictive but also trustworthy.

As the demand for sustainable and ethical investing grows, quantitative finance is also expanding into the domain of ESG—environmental, social, and governance—analysis. ESG investing presents unique challenges, as much of the data involved is qualitative, unstructured, or ordinal in nature. Here, statistics plays a crucial role in developing methodologies for handling such data. Factor models are used to extract signals from ESG scores, while sentiment analysis and natural language processing, grounded in statistical linguistics, are employed to evaluate disclosures and news reports. Bayesian hierarchical models, multi-level regression, and causal inference techniques help quantify the long-term effects of ESG performance on financial returns, allowing for a more robust and principled approach to sustainable finance.

Derivatives pricing, another pillar of quantitative finance, is undergoing a significant evolution with the introduction of models that capture features such as stochastic volatility, jumps, and long memory. Traditional models like Black-Scholes have given way to more flexible frameworks such as the Heston model, rough volatility models, and local-stochastic volatility hybrids. These models are mathematically complex and demand sophisticated estimation techniques. Statistical methods such as the Expectation-Maximization algorithm, Markov Chain Monte Carlo (MCMC), and quasi-likelihood approaches are essential for parameter estimation. Moreover, simulation techniques—especially Monte Carlo and its refined versions—remain indispensable for pricing complex instruments and computing sensitivities. The

calibration of such models to market data requires minimizing error metrics like mean absolute percentage error (MAPE) and root mean square error (RMSE), reaffirming the central role of statistical optimization.

Looking further ahead, quantum computing holds the promise of revolutionizing financial computations. Although still in its infancy, quantum finance seeks to apply quantum algorithms to problems such as portfolio optimization, risk analysis, and option pricing. These applications often involve reimagining probabilistic concepts in a quantum framework, where traditional statistical tools are adapted to handle quantum probability measures and stochastic matrices. As quantum computing matures, the need for statisticians who understand both classical and quantum inference will grow, opening a new interdisciplinary frontier.

In parallel, the study of market microstructure and high-frequency trading continues to gain importance. In this ultra-fast domain, understanding the dynamics of limit order books, execution algorithms, and liquidity provision requires detailed stochastic modeling. Statistical tools such as point process models, particularly Hawkes processes, are increasingly used to capture the self-exciting nature of trade arrivals and order flow. The granularity of high-frequency data enables real-time statistical learning, but also demands precision in estimation and anomaly detection. As algorithmic trading strategies evolve, real-time risk management will rely heavily on adaptive statistical techniques capable of detecting shifts in market regimes.

In conclusion, the future of quantitative finance is deeply intertwined with the evolution of statistics. As models become more complex and data becomes more abundant and diverse, the importance of statistical reasoning only increases. Whether through validating machine learning algorithms, estimating parameters in stochastic differential equations, analyzing unstructured ESG data, or interpreting the probabilistic implications of quantum computations, statistics provides the language, structure, and discipline that the future of finance demands. The field is no longer confined to pricing options or constructing portfolios; it is expanding into domains that require statistical fluency as a precondition for innovation. As such, statisticians and quantitative thinkers are not just contributors to this future—they are its architects.

10 Appendix

10.1 Python Codes

<https://colab.research.google.com/drive/1rd2CdNfRExKXo6a2s2s-eL0cJvwPCyYN>

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